

BASIC COMPUTER SKILLS 3.0

TEACHING MANUAL

CLASS VIII



STATE INSTITUTE OF EDUCATION

DEPARTMENT OF EDUCATION
ANDAMAN & NICOBAR ISLANDS



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MESSAGE

In today's digital age, possessing basic computer skills is no longer a luxury but a necessity. Computers have become an integral part of our personal and professional lives, opening doors to countless opportunities. From communication and research to job applications and online transactions, computer literacy empowers us to navigate the modern world with confidence.

This book on Basic Computer Skills aims to provide the students with a comprehensive understanding of essential concepts, tools, and techniques. This book will guide the students to have a step by step knowledge about the basics of Computer, even if they are a beginner. They will learn about operating systems, word processing, spreadsheets, email communication, internet browsing, and more.

By mastering these skills, they'll not only increase their efficiency and productivity but also unlock new avenues for growth and personal development. I am sure that the introduction of this manual will enable students to embrace the world of technology and empower themselves for success in the digital realm.

I believe that this Teaching Manual will serve as a pivotal moment in revolutionizing the way we educate Middle Level students about computers, and will make them stronger and more knowledgeable in the technical field.

(PallaviSarkar)
Secretary (Education)

ADITYA KUMAR JHA
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FOREWORD

The Department of Education, Andaman and Nicobar Islands in collaboration with SamagraShiksha, UT Mission Authority has been successful in providing ICT Lab facility to Government schools at Middle & Secondary Level throughout the islands with proper ICT facilities along with Internet facility and provision for ICT Instructor.

The students at the Middle & Secondary Level were unable to make use of these facilities to the maximum due to the non-availability of a proper curriculum and syllabus to be taught at different levels. With the objective to have a well-designed structure to introduce the Computer Aided Learning through ICT based skills, the State Institute of Education with the help of a team of Computer Teachers and subject experts have developed a Teaching Manual for the students at Middle Level (Class VI – VIII). The CBSE has also given an outline of the curriculum for IT as a Skill Subject at the Middle Level. The content of the course has also been designed as such to make students capable of independently working on a desktop and be able to develop applications to handle computations of small scale and record keeping. Students will develop the following skills: - Computational and Thinking Skills - Problem Solving Skills - Decision Making - logical and analytical skills.

This course will not only equip students with skills to gain knowledge about the basics of computer but will also instill confidence in them for developing small applications.

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UNIT 1

COMPUTER SOFTWARE AND PROGRAMMING CONCEPTS

INTRODUCTION

In the previous class we studied about the computer storage, how to assemble the parts of computer, generation and types of computer.

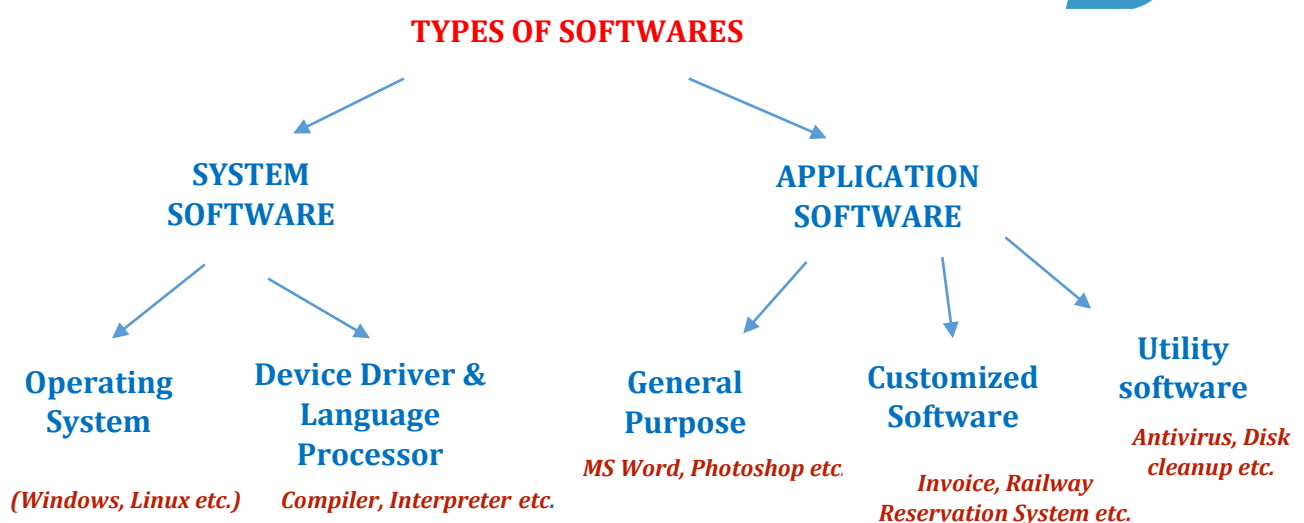


SOFTWARE CONCEPTS

Software is a set of instructions, data or programs used to operate computers and execute specific tasks. It is the opposite of hardware, which describes the physical aspects of a computer. Software is a generic term used to refer to applications, scripts and programs that run on a device.

TYPES OF SOFTWARE

1. System Software
2. Application Software



System software: Software that directly operates the computer hardware to provide basic functionality needed by users and other software and to provide a platform for running application software. System software includes:



- ❖ **Operating System (OS):** Operating system manages resources of computer system like memory, CPU, hard disk, printer etc. also provides an interface between user and computer system & provides various services to other software.
- ❖ **Language Processor & Device Drivers:** All the devices like mouse, keyboard, modem etc. needs at least one corresponding device driver. A device driver is a program that controls a device. A language processor is a system software designed or used to perform tasks, such as processing program code to machine code. Language processors are found in languages such as Fortran and COBOL.

Application software: Software that performs special functions or provides entertainment functions beyond the basic operation of the computer itself. There are different types of application software.

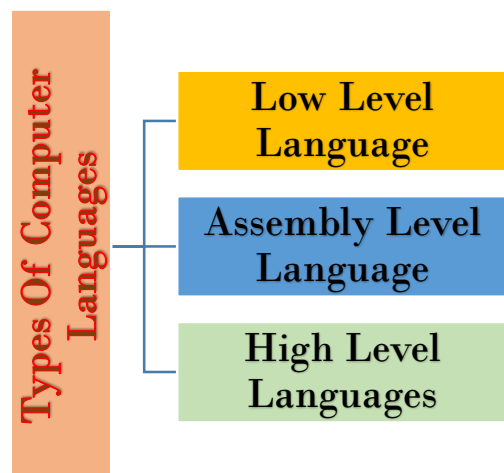
- ❖ **General purpose:** Microsoft Word, Microsoft Excel, MS PowerPoint, Photoshop etc.
- ❖ **Customized:** Invoice Management System, Airline Reservation System etc.
- ❖ **Utilities:** Antivirus, Memory tester, Disk partitioning and Disk defragmenter etc.

COMPUTER LANGUAGES



As we know, to communicate with a person, we need a specific language, similarly to communicate with computers, programmers also need a language which is called as Programming language.

There are three types of Computer Language:



1. Machine Language

Machine language is a type of low-level programming language. It is also called as machine code or object code. The advantage of machine language is that it



helps the programmer to execute the programs faster than the high-level programming language.

2. Assembly Language

Assembly language (ASM) is also a type of low-level programming language that is designed for specific processors. It represents the set of instructions in a symbolic and human-understandable form. It uses an assembler to convert the assembly language to machine language.

3. High-level programming language

High-level programming language (HLL) is designed for **developing user-friendly software programs and websites**. This programming language requires a compiler or interpreter to translate the program into machine language.

The main advantage of a high-level language is that it is **easy to read, write, and maintain**.

High-level programming language includes **Python, Java, PHP, C#, C++, Cobol, Perl, Pascal, FORTRAN etc.**

NUMBER SYSTEM:

When we type some letters or words, the computer translates them in numbers as computer can understand only numbers. This system is called Number System.

There are four types of number system

1. Decimal Number System
2. Binary Number System
3. Octal Number System
4. Hexadecimal Number System

In this unit we shall learn about Decimal and Binary number systems.

Decimal Number System:

The number system that is used in our daily life is the Decimal Number System. μ The decimal number system contains 10 digits. μ The digits used



are 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9. In the decimal number system, the value of a digit depends on its position from the decimal point.

For example, in the number 356, The digit 6 is in the units (100) position. The digit 5 is in the tens (101) position. The digit 3 is in the hundreds (102) position.

Base of a Decimal Number System

The base of a number system is defined as the total number of digits used by that number system. As the decimal number system uses 10 digits, it uses a base of 10. The base of a number system is always written as a subscript of that number. For example, 2310 and 56710 are decimal numbers.



Binary Number System:

Binary Number System is a method of representing numbers that counts by using combinations of only two numerals: zero (0) and one (1). The base of binary number system is 2.

Conversion:

Here is a small table for fast conversion between decimal and binary number systems for the first 16 numbers 0-15:

Binary	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Binary to Decimal Conversion Examples

Convert the binary number 101101_2 to a decimal number.

$$\begin{array}{cccccc}
 1 & 0 & 1 & 1 & 0 & 1 \\
 \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\
 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\
 \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\
 32 & 16 & 8 & 4 & 2 & 1 \\
 \hline
 32 + 16 + 8 + 4 + 2 + 1 = 45
 \end{array}$$

$$101101_2 = 45_{10}$$



Advantages of the Binary Number System

- ❖ Binary number system can easily be converted into other number system & it is the only number system that is understandable by computer

ASSESSMENT

I. Fill in the Blanks

1.is a set of instructions, data or programs used to operate computers and execute specific tasks.
2. A computer can understand andnumbers
3. The number system that is used by man in daily life is thenumber system.
4. The number system is used commonly by computers.
5. language is based upon the objects.

II. Convert the following binary numbers to decimal numbers.

- a. 1110111_2
- b. 1000111_2
- c. 1101100_2
- d. 1011101_2
- e. 1101110_2





UNIT 2

INTRODUCTION TO PROBLEM SOLVING

Learning Objectives:

At the end of this chapter, the students will be able to -

- ❖ Understand the concept of Problem Solving
- ❖ Define the usage of algorithms
- ❖ Identify the various symbols used in Flow charts
- ❖ Draw simple Flow Charts for Mathematical problems

INTRODUCTION

Today, computers are all around us. We use them for doing various tasks in a faster and more accurate manner. For example, using a computer or smartphone, we can book train tickets online.



Computers are used for solving various day-to-day problems and thus problem solving is an essential skill that a student should know. Computers themselves cannot solve a problem. Precise step-by-step instructions should be given by us to solve the problem.

Thus, the success of a computer in solving a problem depends on how correctly we define the problem, design a solution (algorithm) and implement the solution (program) using a programming language.

Thus, problem solving is the process of identifying a problem, developing an algorithm for the identified problem and finally implementing the algorithm to develop a computer program.

ALGORITHM

In our day-to-day life we perform activities by following certain sequence of steps.

Examples of activities include getting ready for school, making breakfast, riding a bicycle, wearing a tie, solving a puzzle and so on.

To complete each activity, we follow a sequence of steps.



Following are the steps required for an activity 'riding a bicycle':

- 1) *remove the bicycle from the stand,*
- 2) *sit on the seat of the bicycle,*
- 3) *start peddling,*
- 4) *use breaks whenever needed and*
- 5) *stop on reaching the destination.*

An algorithm is a detailed step-by-step instruction set or formula for solving a problem or completing a task.

Write the Algorithm to solve the following problems:

Question 1: To polish your shoes.

Algorithm

- Step 1:** *Start*
- Step 2:** *Open the shoe polish*
- Step 3:** *Put shoe polish on brush*
- Step 4:** *Polish one shoe*
- Step 5:** *Put shoe polish on brush*
- Step 6:** *Polish the other shoe*
- Step 7:** *Close the shoe polish*
- Step 8:** *Stop*

Question 2: To make tea/coffee

Algorithm

- Step 1:** *Start*
- Step 2:** *Boil water in a saucepan*
- Step 3:** *Add tea to boiling water*
- Step 4:** *Add sugar to boiling water*
- Step 5:** *Add milk to boiling water*
- Step 6:** *Boil this water with all the ingredients for 2 mins*
- Step 7:** *Sieve the tea in a cup*
- Step 8:** *Stop*

FLOWCHART

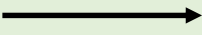
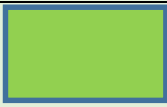
A flowchart is a pictorial representation of an algorithm.

A flowchart is a diagram made up of boxes, diamonds and other shapes, connected by arrows.



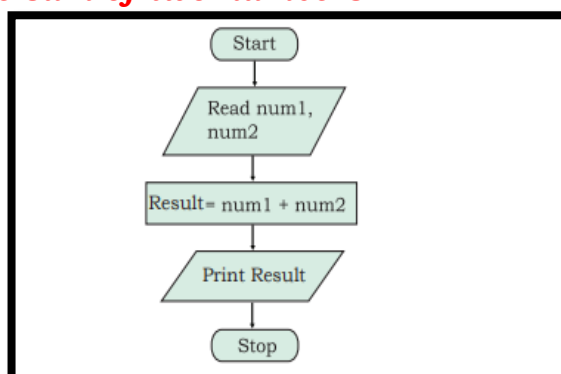
Each shape represents a step of the solution process and the arrow represents the order or link among the steps.

Shapes or symbols to draw flow chart

SYMBOLS	NAME	FUNCTIONS
	Start/End	Oval represents Start or End
	Arrows	Connector to show order of flow between shapes.
	Input/Output	Parallelogram shape is used to input or output data.
	Process	Also called “ <i>Action Symbol</i> ,” it represents a process, action, or a single step.
	Decision	A decision or branching point, usually a yes/no or true/ false question is asked, and based on the answer, the path gets split into two branches.

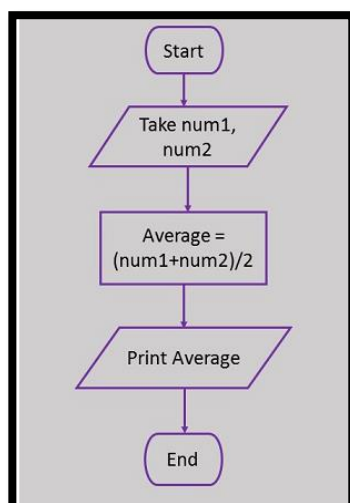
Draw the flowchart for the following:

1. To find the sum of two numbers



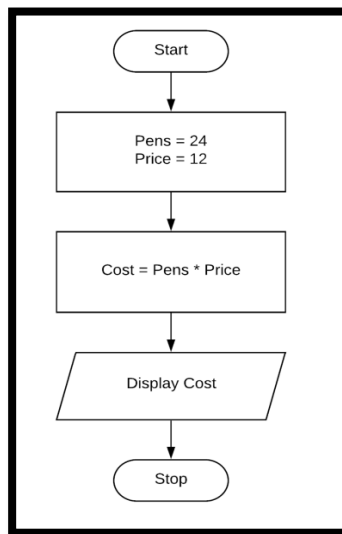
Flowchart to display the sum of two numbers

2. To calculate the average of two numbers.



Flowchart to calculate the average of two numbers

3. To find the cost of 24 pens when the cost of one pen is Rs. 12.



Flowchart to calculate the cost of 24 pens if one pen cost Rs. 12/-

4. Write Algorithm and flowchart to find the cube of a given number.

Algorithm

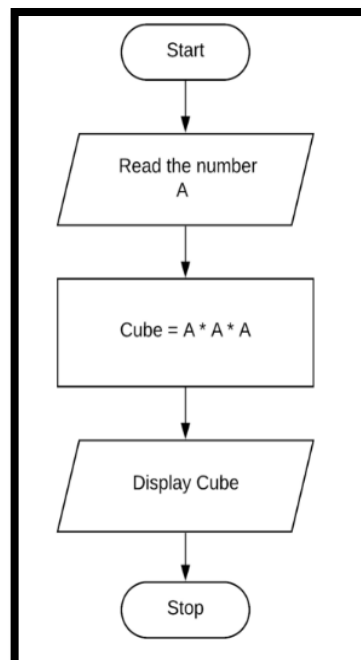
Step 1: Start

Step 2: Read the number and store it in a.

Step 3: Calculate cube of number as $a * a * a$

Step 4: Print cube of number

Step 5: Stop

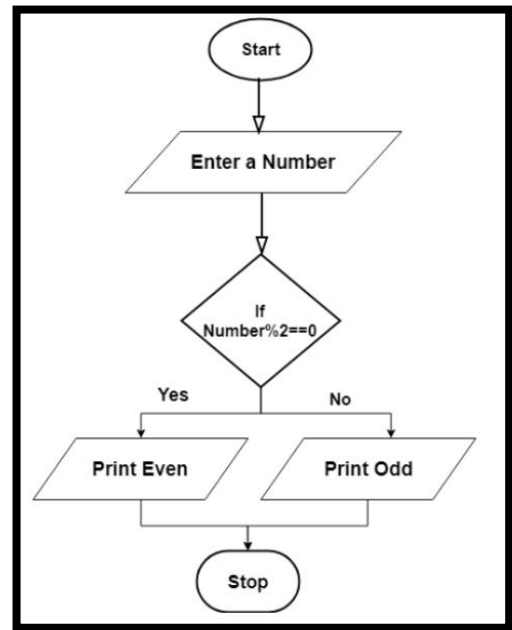


Flowchart to calculate cube of a given number

5. Write Algorithm and flowchart to find a number is even or odd.

Algorithm

- Step 1:** Start
Step 2: Read the number
Step 3: Calculate remainder=Number%2
Step 4: if remainder==0;
Step 5: print even
Step 6: else print odd
Step 7: Stop



Flowchart to find the number is Even or Odd

Test your knowledge

I) Match the following columns Column A Column B

Column A

1. 
2. 
3. 
4. 
5. 

Column B

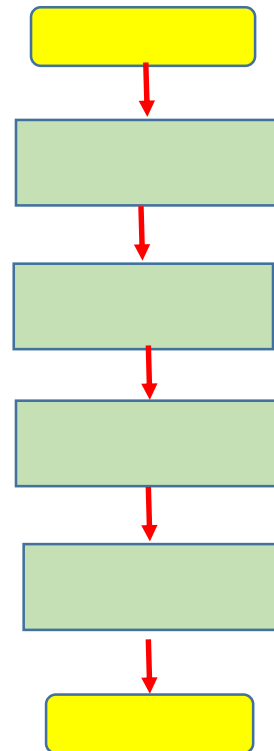
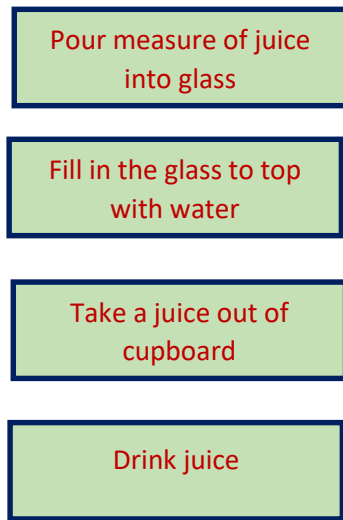
- a) Flow of Control
- b) Process Step
- c) Start/Stop of the Process
- d) Data
- e) Decision Making



II) Pictorially represent the given Algorithm, to add two numbers

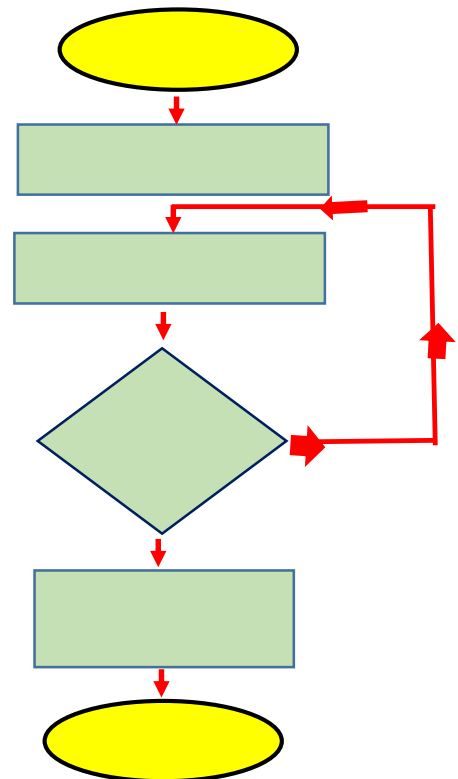
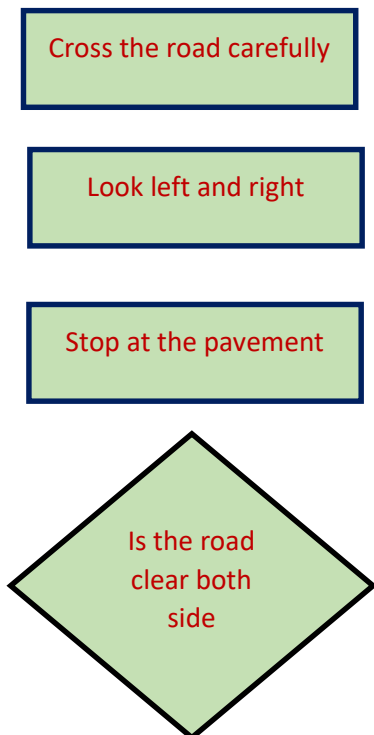
- Step 1:** Start
Step 2: Initialize sum = 0
Step 3: Read num1, num 2
Step 4: Calculate sum = num1+num2
Step 5: Print sum
Step 7: Stop

III) Place the instructions below in the flow chart. Make sure you put the instructions in the correct order.



IV) The flow chart on the right is meant to show the steps for safely crossing the road. There is a decision box in this flow chart.

Place the boxes below in the flow chart.



V) Draw the flow chart to find the greatest among two different numbers entered by the user.



Learning Objectives

- ✓ What is Scratch? Getting familiar with Scratch User Interface
- ✓ Working with Code Blocks and Sprite
- ✓ Script Execution & Adding Sounds
- ✓ Changing Backdrop & Editing Costume
- ✓ Sprite Manipulation
- ✓ Creating own Sprite
- ✓ Conditional Decision Making Statements



A computer also follows stepwise instructions to complete any task known as a program. We can define a program as a set of systematic instructions that direct the computer to do specific tasks and get the desired results. Programs are written in a special language known as a programming language. The person who writes the programs is known as a programmer.

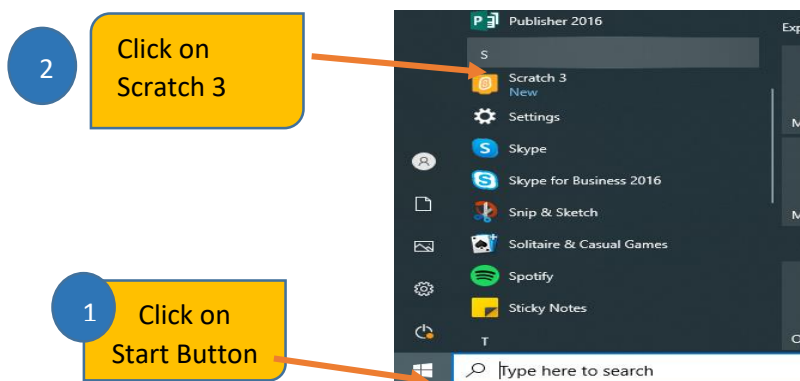


ABOUT SCRATCH

Scratch is a visual programming language used to create computer animations, games, interactive stories and other visually rich programs. It is block based programming language. Here we use blocks of code that we put together to create our projects. It enables us to see the results of our work quickly and even include graphics and sound to it.



GETTING STARTED WITH SCRATCH

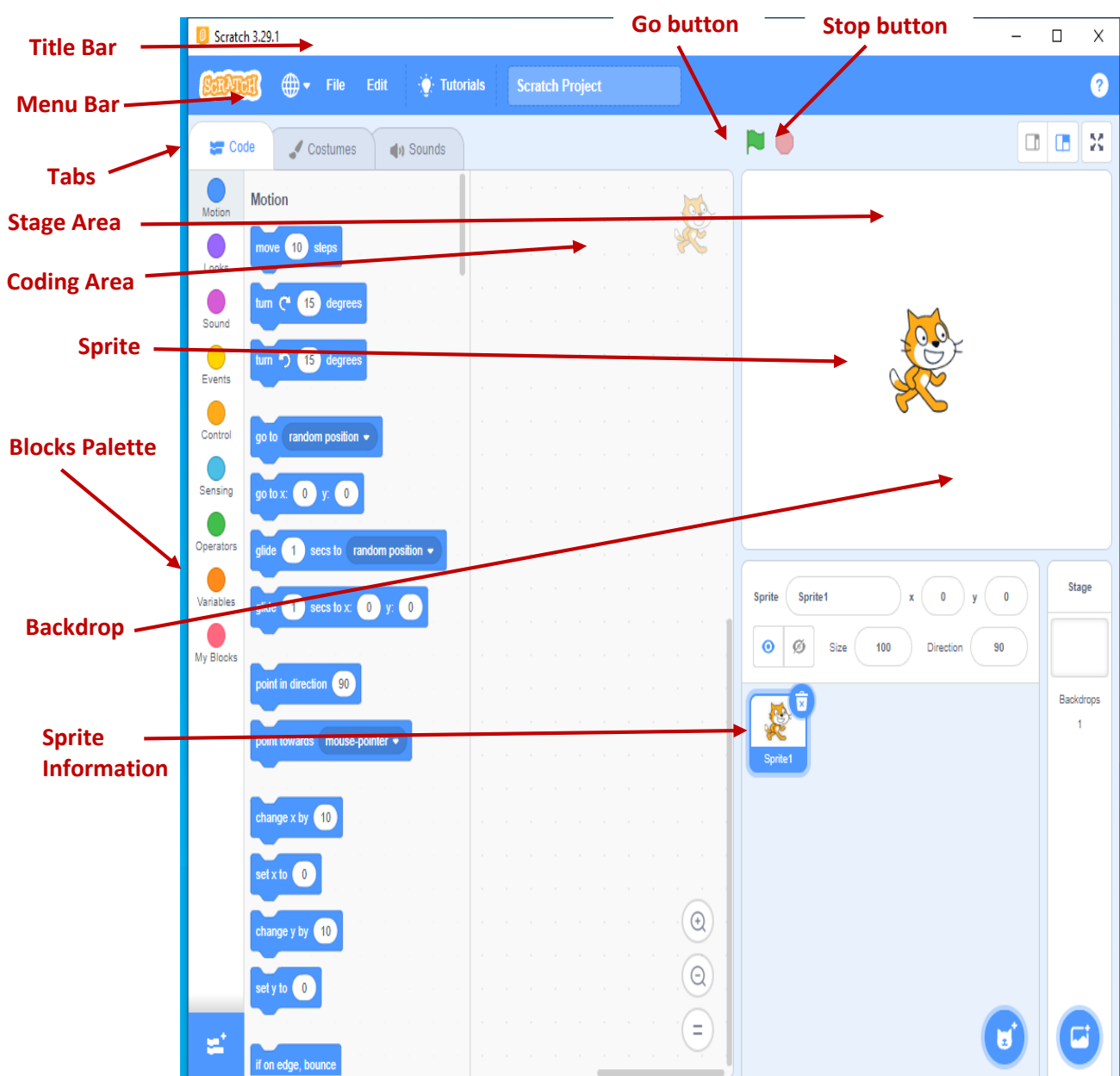


Scratch was created by Mitchel Resnick at the MIT Media Lab in 2007

Info - There are two versions of the Scratch offline editor available:

- Use Scratch Desktop 3.0 If your computer is running Windows 10 or higher, or macOS 10.13 or higher.
- Use Scratch 2 if your computer uses an older operating system.

Scratch is launched, you will see a window as given below-



Title Bar – It is present at the top of the Scratch window. It shows Scratch written on the left. The right side of the Title Bar has the three control button: **Minimize, Maximize and Close**.



- ❖ **Menu Bar** –It contains options and commands related to the projects that you create in Scratch. The options are **File, Edit, Tutorial and name of the Scratch project.**
- ❖ **Tabs** – Scratch has mainly three tabs.
 - **Code Tab** –It contains block categories and list of blocks in a category. Some block categories are: Motion, Look, Sound, Pen, Events, Control, sensing, Operators, Variables and My Blocks. Each block performs a task in a project.
 - **Costumes Tab** – In this tab, you can add and make changes in the appearance of the sprites and backdrops.
 - **Sounds Tab** – In this tab, you can add and make changes in the sounds of the sprite and backdrop.
- ❖ **Sprite** – It is a character or object in Scratch which acts on the stage. An orange cat is the default sprite.
- ❖ **Coding Area** – It is a place where you drag blocks and join them to create your code.
- ❖ **Stage Area** – It is a place where you will see all the action happening.
- ❖ **Backdrop** – It is a background of the stage area.
- ❖ **Sprite Information Pane** – It shows all the details - sprites and backdrops used in a project. You can add, delete, see the name and location, increase and decrease the size of a sprite. Similarly, you can add or change the backdrop.
- ❖ **Block Palette** - The block palette consists of 9 different script blocks. The different type of script blocks include: Motion, Control, Looks, Sense, Sound etc.
- ❖ **Add Extension** – It takes us to menus of extensions from where we can add extra blocks to the code area.
- ❖ **Go Button** – It is the green flag and when you click on it, it runs a Scratch program.
- ❖ **Stop Button** – It is the red button which on clicking stops the running Scratch program.



WORKING WITH CODING

Coding blocks are used to create a code. Different types of blocks have different colors. There are total 9 such blocks. They are:

MOTION BLOCK

These blocks are used to control the movement of sprites. These blocks are dark blue in color and have 18 blocks under them.

LOOKS BLOCK

These blocks are used to control the appearance of the sprite. They are purple in color and have 20 blocks under them.

SOUND BLOCK

These blocks are used to control the sound functions. They are dark pink in color and have 9 blocks under them.

EVENTS BLOCK

These blocks are used to manage the events which trigger the movement of the sprite. They are yellow in color and have 8 blocks under them.

CONTROL BLOCK

These blocks are used to control the scripts under some specific conditions. These blocks are amber in color and have 11 blocks under them.

SENSING BLOCK

These blocks are used to sense or detect different factors of a project. They are light blue or cyan in color and have 18 blocks under them.

OPERATORS BLOCK

These blocks are used to perform mathematical operations and string handling. They are green in color and have 18 blocks under them.

VARIABLES BLOCK

These blocks are used to report the value of a variable. They are orange in color and have 5 blocks under them.

There is a sub-category of blocks under Variable blocks known as List Blocks. List Blocks are used to manipulate different values in Scratch. These blocks are dark red in color and have 12 blocks under them.

MY BLOCKS

This block is used to hold the procedure of selected sprites. They are pink in color and have a “Make a Block” button using which customized blocks can be created.



WORKING WITH SPRITE



In Scratch, any character or object is called a sprite. Every new project in Scratch starts with the Cat sprite.

Do you want to choose a different sprite? Then follow these steps:

1 Click the New Sprite icon

2 Click on the sprite that you like.

3 Your Sprite is added

4 Click here to Delete a sprite



CONTROLLING SCRIPT EXECUTION AND ADDING SOUNDS

CREATING A SCRATCH PROJECT

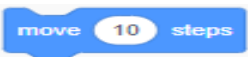
Project 1: Move a Sprite and code it to say “Hello”.

Let's see how a sprite can be moved a few steps ahead and say “Hello”.

Step 1: Click on the **Event Block**.

Step 2: Drag  a block into the Code Area.

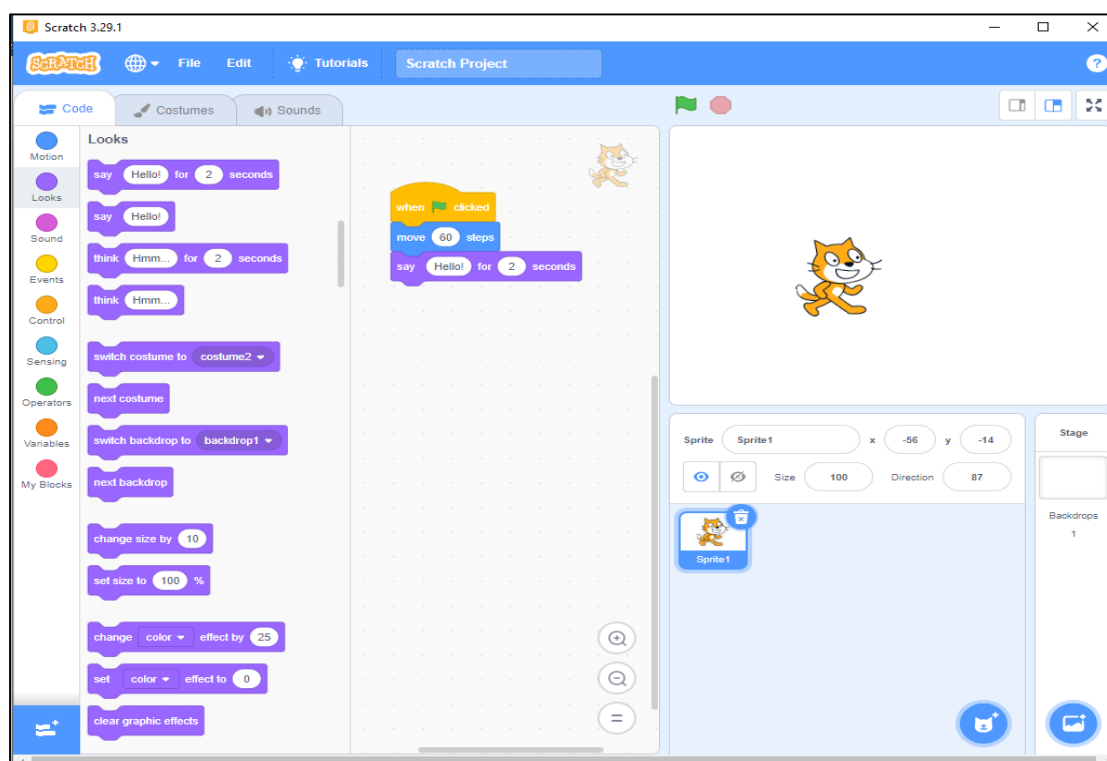
Step 3: Click on the **Motion Block**.

Step 4: Drag a  block.

Step 5: Click on the number 10 and change it to  (The number determines how far across the screen you want the sprite to move.)

Step 6: Click on the **Look Block** and drag a block. This ‘Say’ block has two inputs: one for what will be displayed in the speech bubble, and one for the number of  seconds that the speech bubble will show on the **Stage**.

Step 7: Click the (Go Button) to run your program.



Save your project

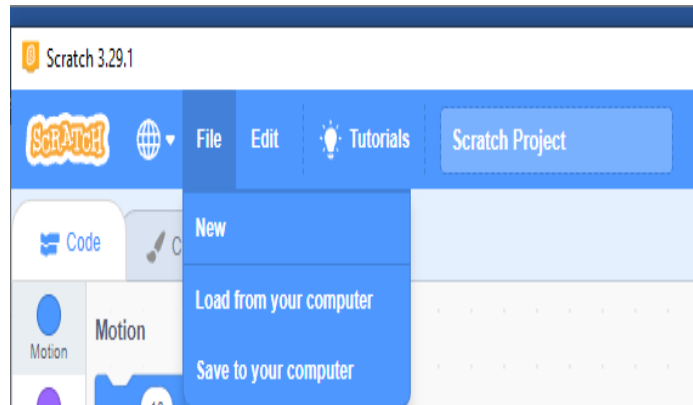
Click the File menu and select **Save to your computer**.

Open your Existing project

To open a project, click **Load from your computer**.

Open a New project

Click the File menu and select **New**.



Project 2: Adding Sound.

Let's add meow sound to a cat sprite.

Step 1: Click on the **Event Block**.

Step 2: Drag a **when clicked** block into the Code Area.

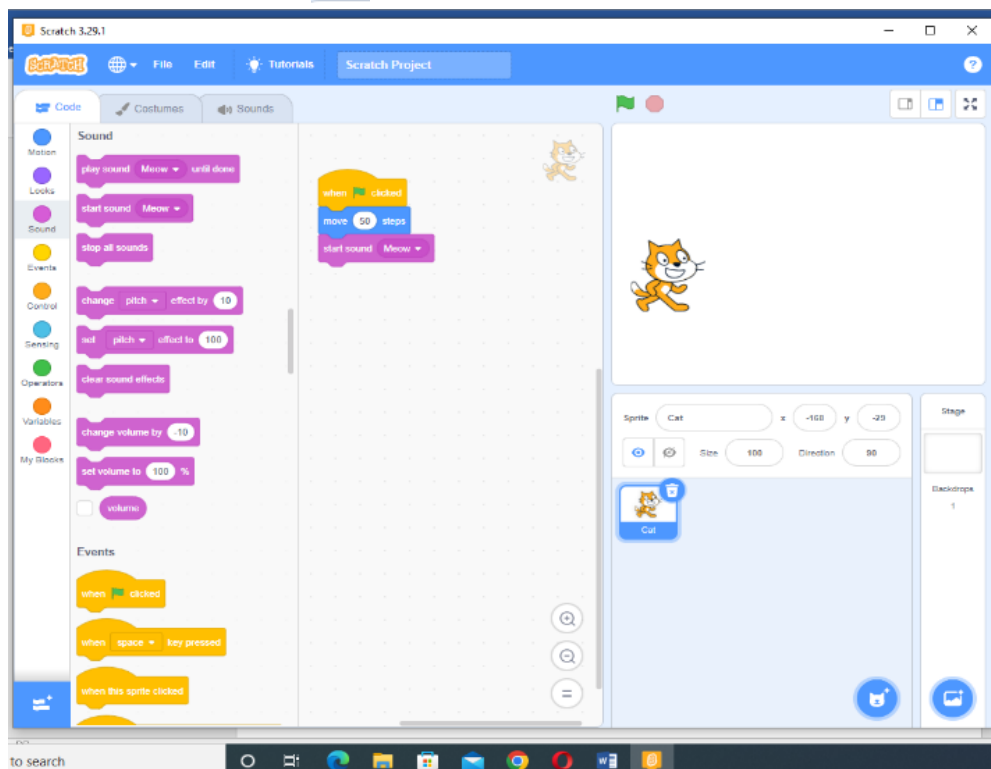
Step 3: Click on the **Motion Block**.

Step 4: Drag a **move 60 steps** block.

Step 5: Click on the **Sound Block**.

Step 6: Drag a **start sound Meow** block.

Step 7: Click the **Go Button** (Green Flag) to run your program.





CHANGING BACKDROP & EDITING COSTUME

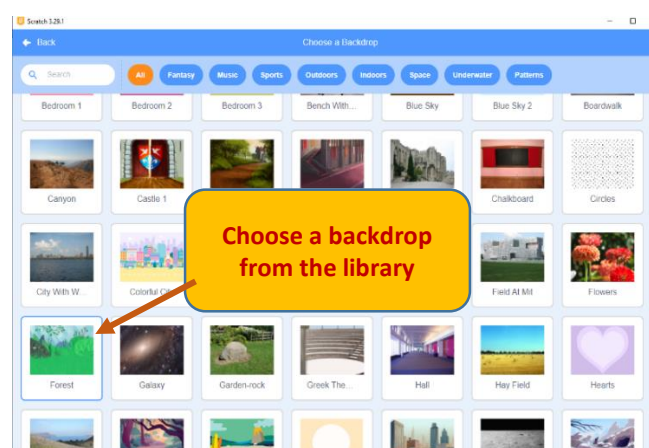
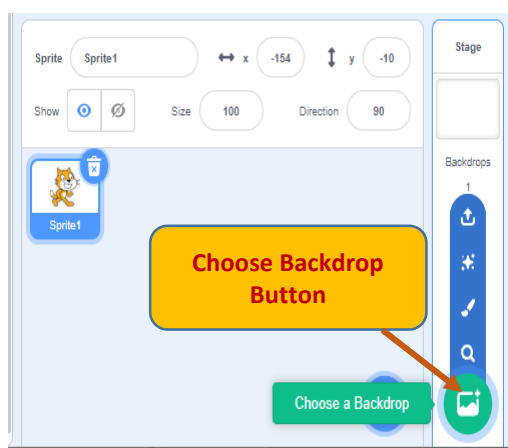
A backdrop is background image that can be shown on the stage area (by default, a white backdrop appears on the stage area).

To use a backdrop in Scratch, follow these steps:

Step 1: Open Scratch and click on the “Choose a Backdrop” option which is in blue color in the shape of a circle available at the extreme bottom left corner of the Scratch user interface.

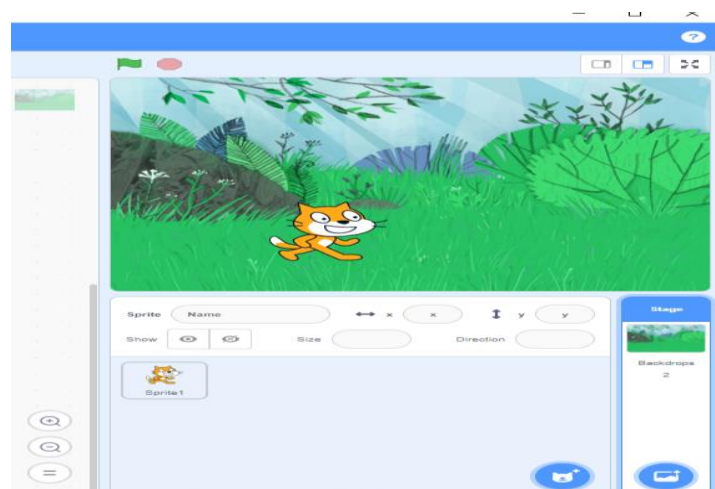
Step 2: Once you click on the blue button, the following options will appear on the screen. You can choose, paint, surprise, and upload a backdrop using these options.

Step 3: To choose a particular backdrop, click on the first option that says, “Choose a Backdrop”.



Step 4: Select the backdrop from the library.

Step 5: A new background appears on the stage. The cat sprite will be in front of your background.



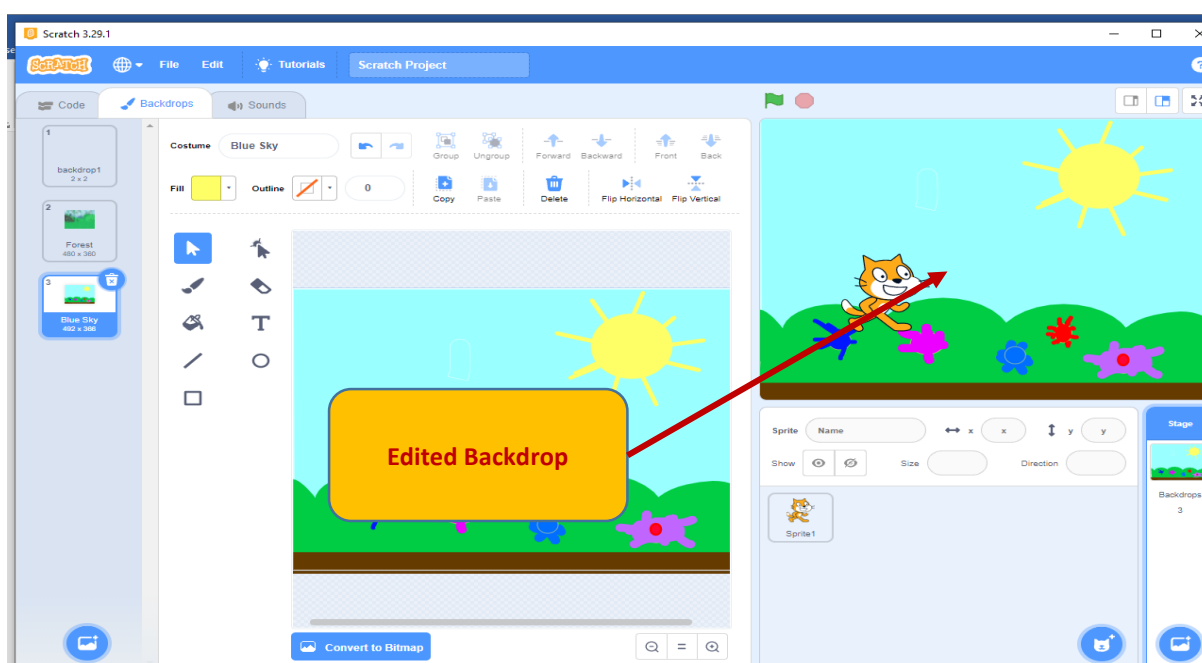
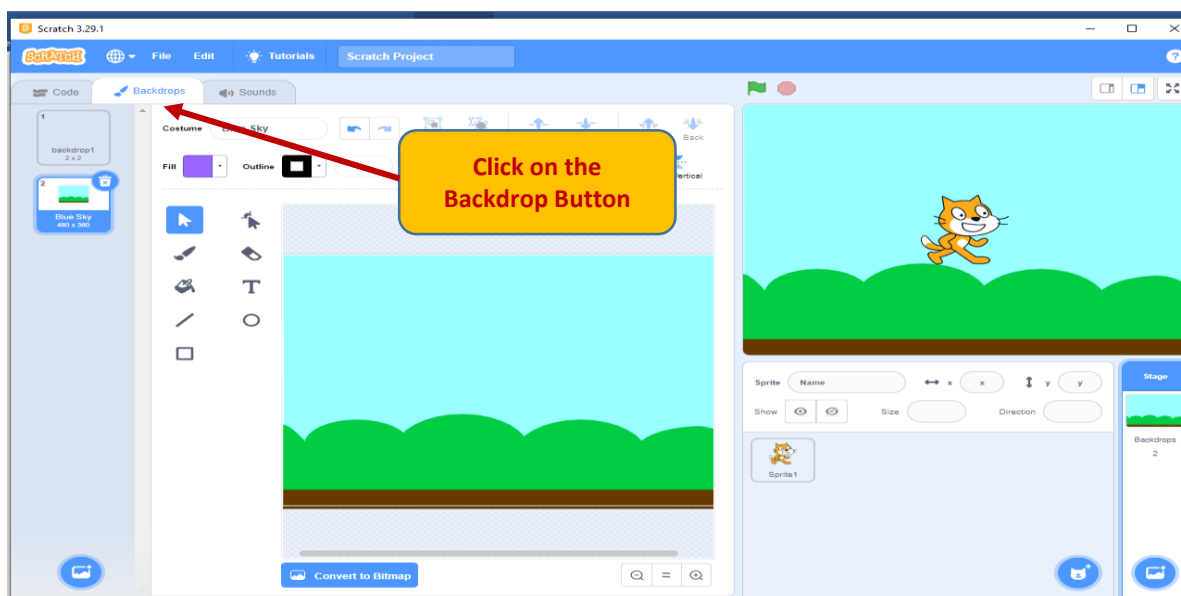


To edit and delete a backdrop in Scratch, follow these steps:

Step 1: Click on “Backdrop Tab”.

Step 2: Now you can edit the backdrop using the available tools present on the left side of the editing screen.

Step 3: To delete a backdrop, select the backdrop you want to delete from Backdrop Tab. Then click on delete option.





CHANGING THE COSTUMES OF A SPRITE

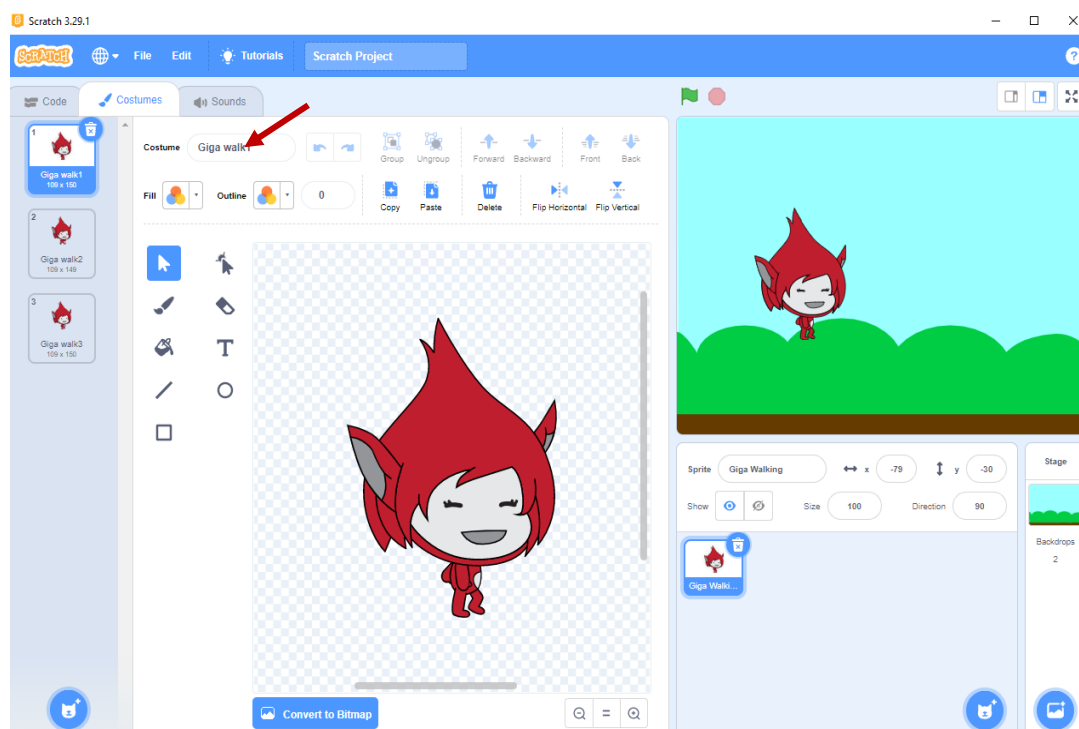
Costume is a method to dress up a character or a sprite. You can add different accessories to make sprite look better. These costumes can be named, created, edited, deleted, and changed but every sprite must have at least one costume.

To change a costume in Scratch, you can do the following steps:

Step 1: Open scratch and click on the “Costumes” option available at the top of the screen.

Step 2: Once you click on the “Costumes” option, the following screen will appear.

Step 3: Now, here using the options available, you can change the costume in Scratch.



SPRITE MANIPULATION, CREATING OWN SPRITES



Every new project starts with one sprite (the cat) already loaded, but you can also add as many of your own as you want, or you can edit or delete the cat.

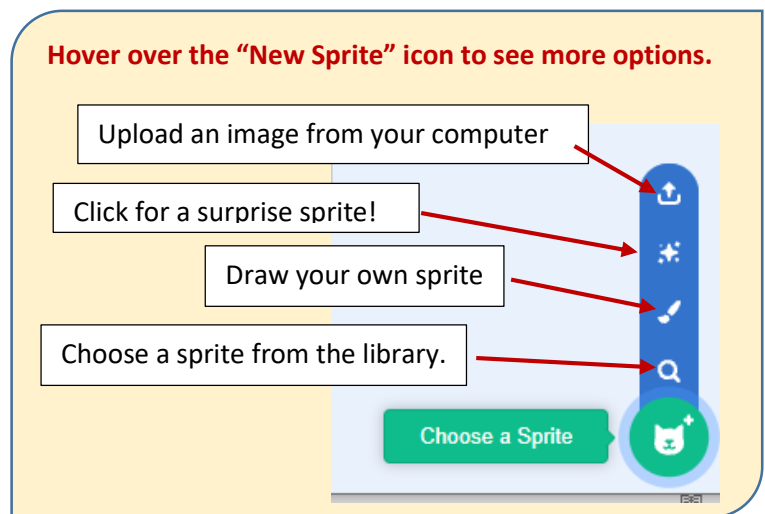
CREATING SPRITE

As with backdrops, there are four icons for creating a sprite:

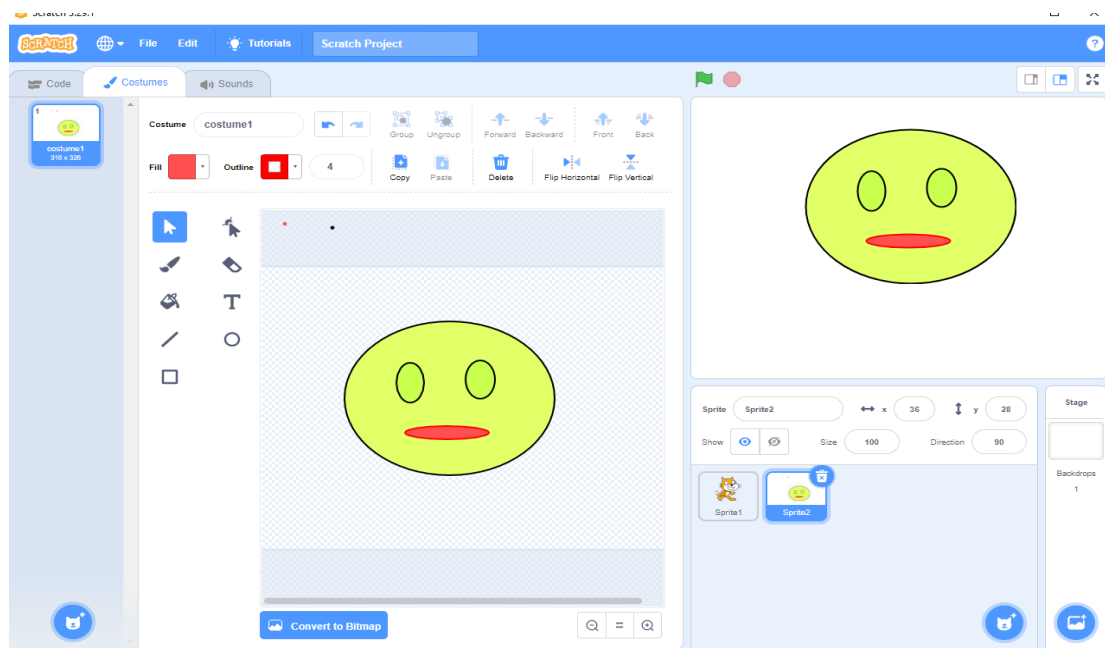
- Choose sprite from library
- Paint new sprite
- Upload sprite from file
- New sprite from camera

Choose sprite from library:

Using this option you can choose the sprite available in the library.



Paint new sprite: This option can be used to create or edit your sprite.



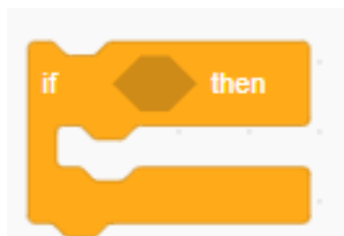
CONDITIONAL DECISION MAKING STATEMENTS

Conditional statements have slots that are shaped with points on either side which evaluate to a true or a false value and execute if the statement is true. They are found in the **controls** programming blocks and are used for program flow with **if**, **repeat**, **forever**, and **wait** blocks. The conditional part of the statement is found amongst the **sensing** and the **operator** areas.

CONDITIONS

In programming, a condition is something that must be true in order for something to happen. A condition is thus said to “evaluate to true” or “evaluate to false.” In Scratch, any block whose label says “if,” “when,” or “until” is a sort of conditional construct.

One such block is:



The construct above is generally known as an “if-then construct.” With it we can instruct a sprite to say hello only if, the user has pressed the mouse button:



A related construct is the “if-then-else construct”:

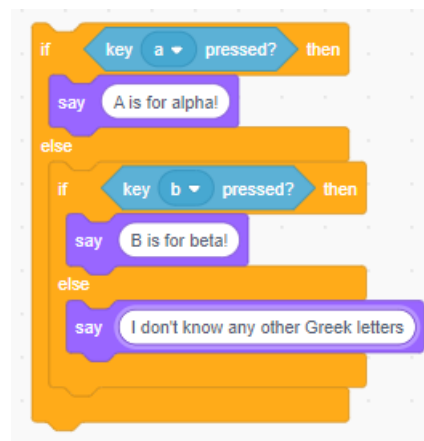


With the above construct we can instruct a sprite to say hello or goodbye, depending on whether the user has pressed the mouse button:



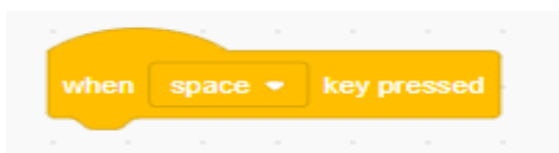


The above constructs can be nested to allow, for example, for three different conditions:



The above construct could be called an “if-then-else if-then-else construct”.

Another conditional block is:



Yet another such block is:

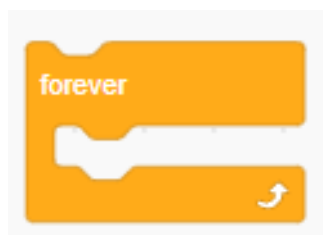


Sometimes, we want one or more statements to be executed multiple times in a row. To implement this behavior, we turn our attention to LOOPS.

LOOPS

In programming, a loop can induce multiple executions of statements. In Scratch, any block whose label begins with “**forever**” or “**repeat**” is a looping construct.

One such block is:

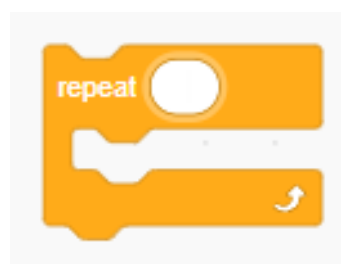




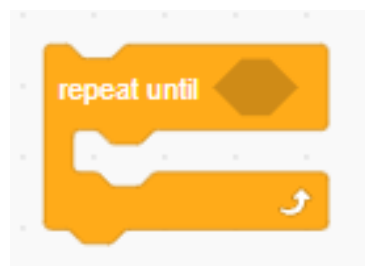
This construct allows us, for instance, to instruct a sprite to meow every other second:



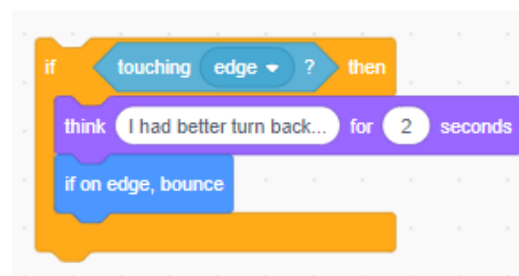
Another block allows us to loop a specific number of times:



And another block allows us to loop until some condition is true:



Conditional statements ask questions about the program state to choose from a set of different sequences of commands. In Scratch for example, you can determine whether you are at the edge of the stage with the **if touching edge** control block.





1. _____ is the default Sprite in Scratch.
2. Scratch was created by _____ at MIT Media Lab.
3. _____ is a character or object in Scratch which acts on the stage.
4. _____ is the red button which on clicking stops the running Scratch program.
5. _____ is a place where you will see all the action happening.

1. MOTION BLOCK	a. Adds sound
2. LOOKS BLOCK	b. To perform mathematical operations
3. SOUND BLOCK	c. To detect different factors of a project.
4. SENSE BLOCK	d. To move sprite
5. OPERATOR BLOCK	e. Purple color

1. **If**, **repeat**, **forever**, and **wait** blocks are found under _____ programming block.
 - a) Sound
 - b) Control
 - c) Event
 - d) Operators
2. In Scratch, any block whose label begins with _____ or repeat is a looping construct.
 - a) When
 - b) If-then
 - c) If-then-else
 - d) Forever
3. _____ number of code blocks are available in Scratch 3.0.
 - a) 6
 - b) 4
 - c) 9
 - d) 10



4. _____ color is used for Sound code block.

- a) Yellow
- b) Dark pink
- c) Green
- d) Orange

5. _____ option is used to change the background of your stage area.

- a) Backdrop
- b) Motion Block
- c) Script
- d) Sprite



UNIT 4

PRESENTATION TOOL (MS-POWER POINT)

Learning objectives:

In this chapter you will be able to:

- ✓ Insert different objects in a slide such as image, audio, video, etc.
- ✓ Add transitions in slide
- ✓ Apply custom animations with text
- ✓ Use shortcuts in Power Point
- ✓ Insert and use various charts in your presentation

INTRODUCTION TO MS-POWERPOINT

PowerPoint is a presentation software program of the Microsoft Office package. As we have already learnt in Class VII about using slide layouts, inserting text, images and other objects in a slide, applying templates and running the slide show.

In this chapter we will discuss some advance features of PowerPoint and its use to prepare attractive presentations.

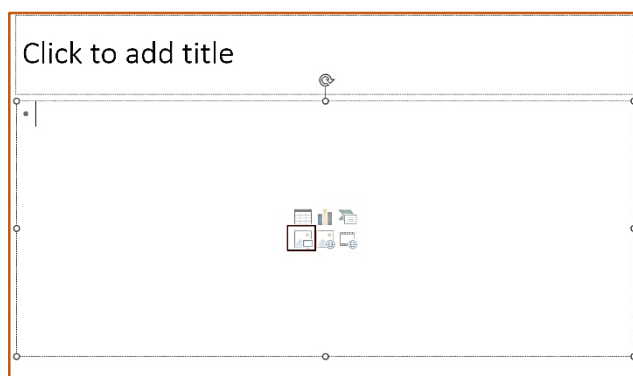
INSERTING DIFFERENT OBJECTS IN POWERPOINT SLIDE

1. Inserting a picture

There are two ways to insert a picture on your slide in PowerPoint:

Using Content Placeholder

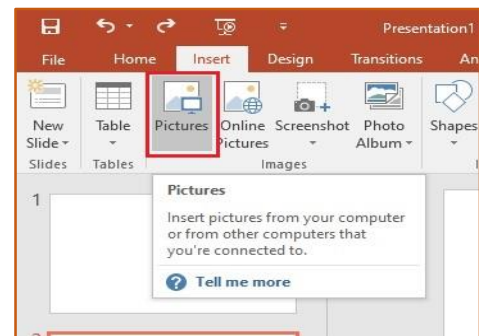
(i) Launch PowerPoint. Most of the time, PowerPoint already has a new slide, change the layout of this slide to Title and Content by selecting the Home tab -> Layout -> Title and Content option.



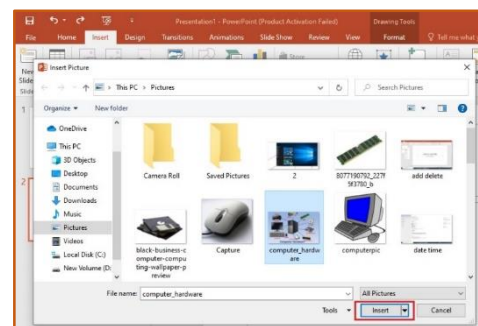
These results in the slide layout you see in the Picture. Click on the **Insert Picture** button in the Content placeholder as highlighted in red.

Using Insert Picture Tool

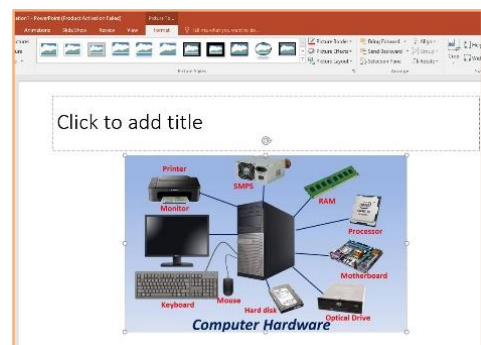
1. If you want to add a picture on an existing slide that has no Content placeholder, you can select the **Insert** tab of the Ribbon, and click the **Picture** button.



2. Once you click on the Picture option, **Insert Picture** dialog box will appear. Navigate to the folder where your pictures are located, you will see the icons or thumbnails of the pictures as shown in following picture. Select the picture you need, and click on the **Insert** button highlighted in red.



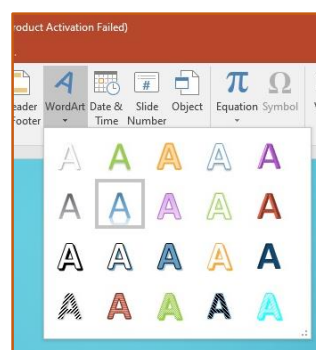
3. As soon as you click the Insert button, you will see the selected picture placed on the PowerPoint slide. Now, the picture can be edited by resizing, moving, rotating, etc.



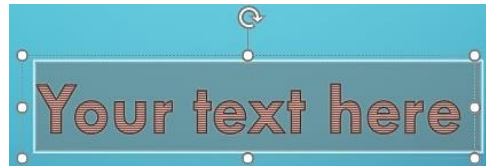
2. Inserting Word Art

WordArt offers additional effects like outline text, fill, gradient, 3D effects and shadows etc.

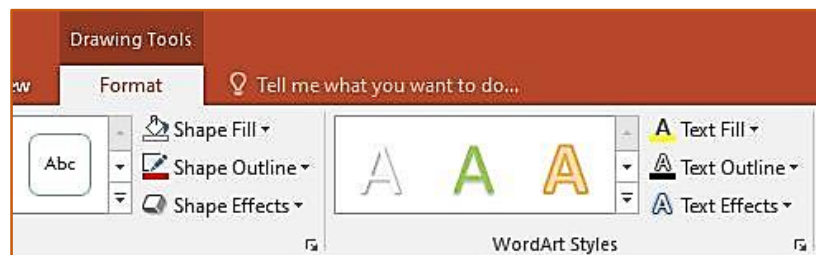
- ❖ On the **Insert** tab, in the **Text** group, click **WordArt**. The WordArt Styles gallery opens.



- ❖ Click on the style you want to insert, and a text box appears with the text “Your text here”, with the WordArt style you selected applied to it.



- ❖ Edit the text of your choice. At this point, you’ll see a **Drawing Tools Format** context tab. Here you’ll find the WordArt Styles group



- ❖ This contains the Preset styles gallery, and drop downs for controlling the Text Fill, Text Outline, and Text Effects.



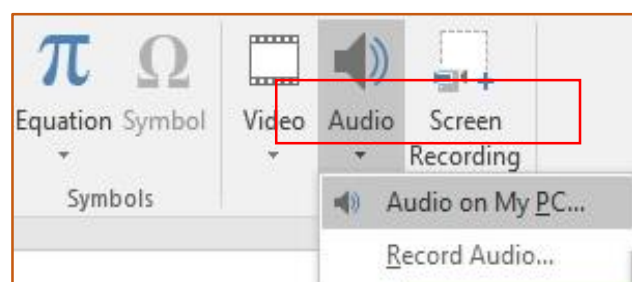
- ❖ Final output will be like the picture shown here.



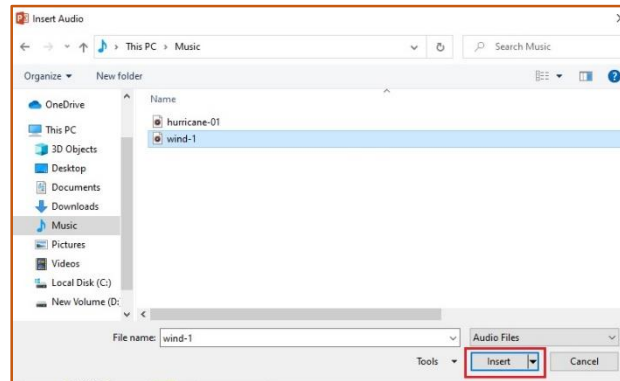
3. Adding Audio to a PowerPoint Presentation

To insert audio from a file:

- ❖ From the **Insert** tab, click the **Audio** drop-down arrow, then select **Audio on My PC**.



- ❖ Locate and select the desired audio file, then click **Insert**.



- ❖ The audio file will be added to the slide.

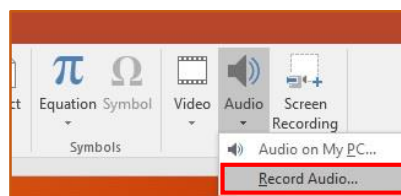


Recording your own audio

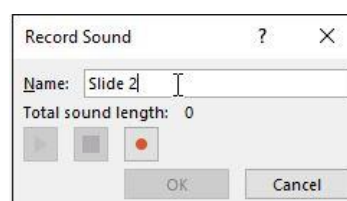
Sometimes you may want to record audio directly into a presentation. For example, you might want the presentation to include narration. Before you begin, make sure you have a microphone that is compatible with your computer; many computers have built-in microphones or ones that can be plugged into the computer.

To record audio:

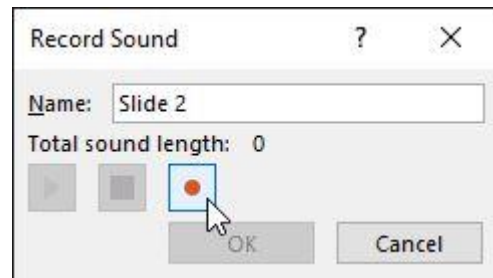
- ❖ From the Insert tab, click the **Audio** drop-down arrow, then select **Record Audio**.



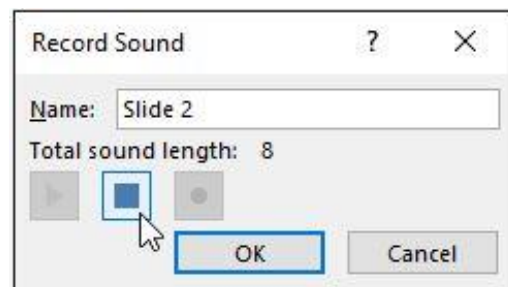
- ❖ Type a suitable name for the audio recording.



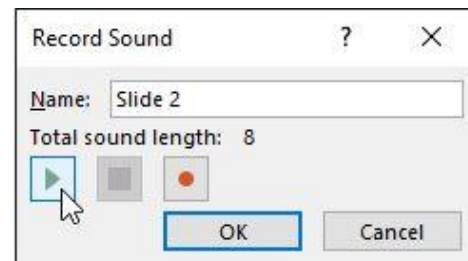
- ❖ Click the **Record** button to start recording.



- ❖ When you're finished recording, click the **Stop** button.



- ❖ To preview your recording, click the **Play** button.



- ❖ When you're done, click **OK**. Your recorded audio file will be inserted into the slide.



To move an audio file:

- ❖ Click and drag to **move** an audio file to a new location on a slide.

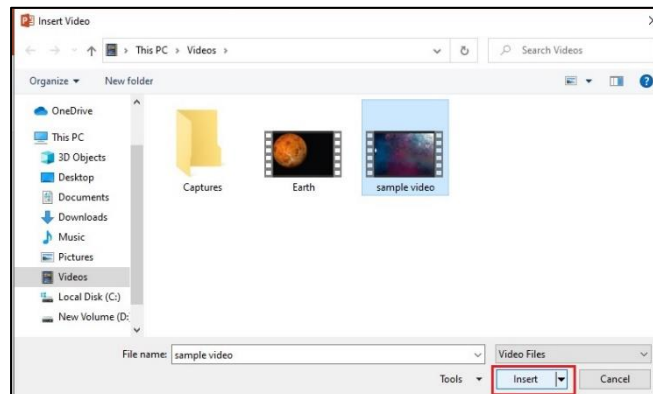
To delete an audio file:

- ❖ Select the audio file you want to delete, then press the **Backspace** or **Delete** key on your keyboard.

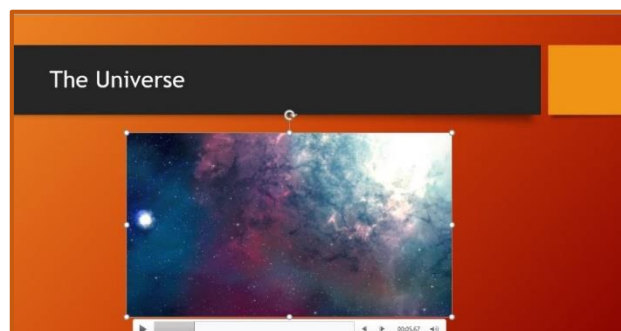
4. Inserting Video to a PowerPoint Presentation

To insert a video file into PowerPoint:

- ❖ Open your presentation and select a slide. From the ribbon bar, click **Insert > Video > Video on my PC**.
- ❖ Locate the video file on your PC in the **Insert Video** box, then click the **Insert** button to insert it into your presentation.

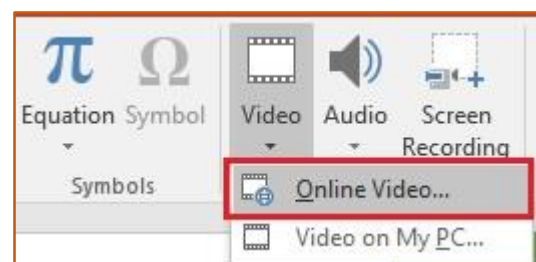


- ❖ Your video file will get inserted into your slide.

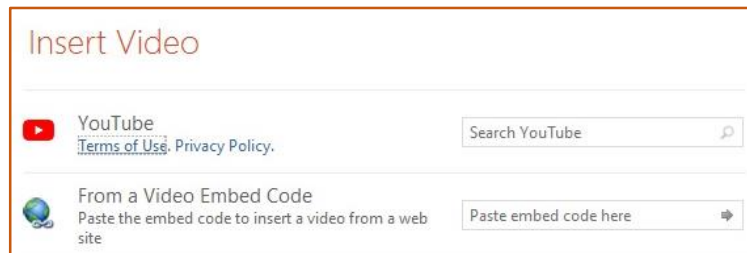


Embedding YouTube Videos to a PowerPoint Presentation

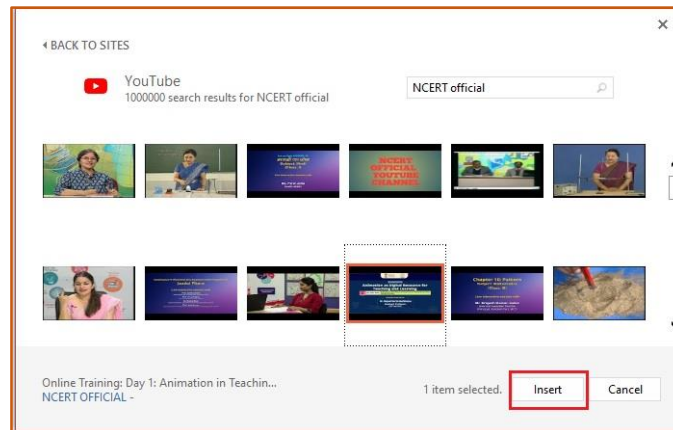
- ❖ You can also insert online videos. From the ribbon bar, click **Insert > Video > Online Video**.



- ❖ In the **Insert Video** dialog box, search for the YouTube video you want to insert.



- ❖ Select the video and click **Insert** to add the video to your presentation.



To move a video file:

- ❖ Click and drag to **move** the video file to a new location on a slide.

To delete a video file:

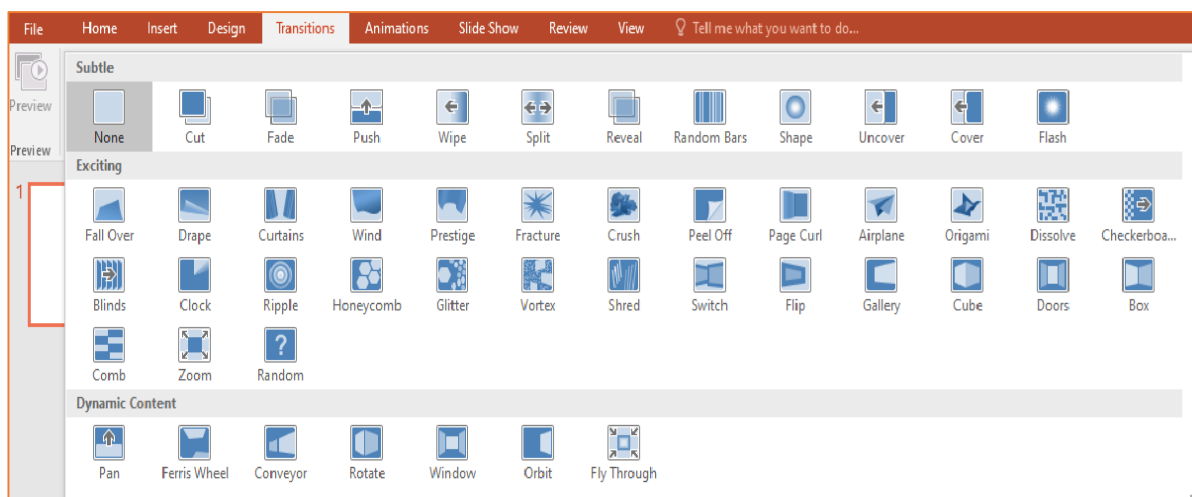
- ❖ Select the video file you want to delete, then press the **Backspace** or **Delete** key on your keyboard.

TRANSITIONS IN SLIDE

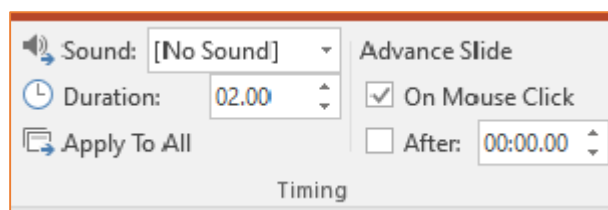
Slide transition is the animation that occurs when you advance from slide to slide within the presentation. You have many options for controlling the effects of the slide transition animation in PowerPoint.

For example, you can often set the speed and direction of many transition animations. You can also add sound effects to complement the slide transition animation in PowerPoint

Applying Transition to the Slides:



- Select the slide you want to modify.
- Click on **Transition** tab.
- Select the transition of your choice and apply it to the selected slide.
- Click on the **Sound** dropdown and select sound effect for the slide.
- Set a duration for the slide transition from **Duration** option.
- In the **Advance Slide** option select either **On Mouse Click** or **After** options. For After option set the duration.
- Click **Apply to All**.



CUSTOM ANIMATIONS WITH TEXT

Custom animation is a set of effects which can be applied to objects in PowerPoint so that they will animate in the Slide Show.

PowerPoint provides four types of animations. They are:

- **Entrance animation:** An entrance animation determines the manner in which an object appears on a slide or move onto a slide.
- **Emphasis animation:** An emphasis animation does something to draw the attention of the audience towards the object.

For example, the object is enlarged.

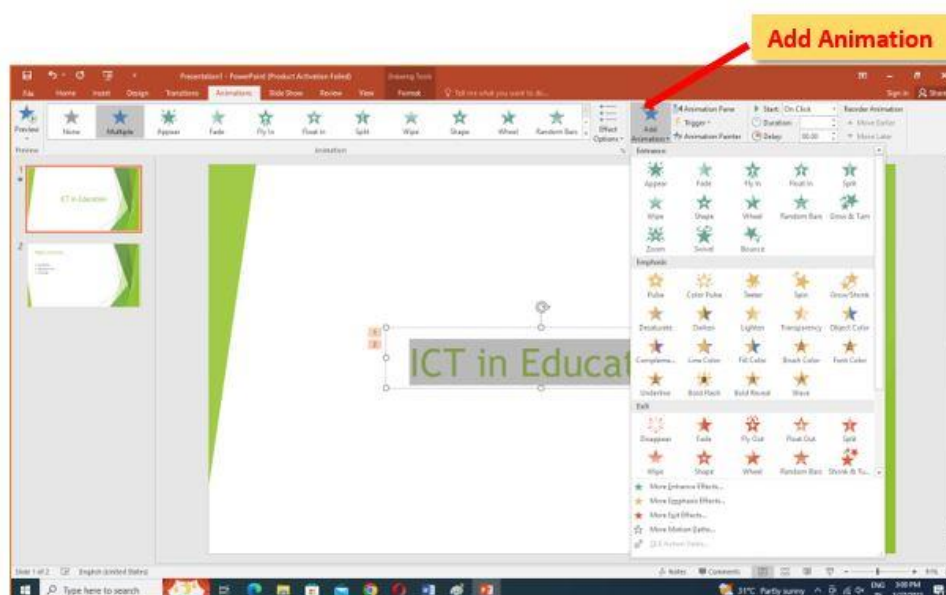


- **Exit animation:** An exit animation determines the manner in which an object leaves a slide.
- **Motion paths animation:** This animation determines how an object moves around in a slide

Applying Custom Animation in a slide:

Select the text to be animated on the slide.

- Click on **Animations** tab.
- Click on **Add Animation** and choose from the drop-down menu.
- Select the effect you want to apply to your text. Your added effect on text is reflected on the slide show.



DID YOU KNOW?

PowerPoint was originally called "Presenter". It was created in 1987 by a company called Forethought Inc. for Macintosh.

SHORTCUT KEYS IN POWERPOINT

PowerPoint shortcut keys help users to use the functions of PowerPoint with more efficiency. They allow users to keep hands on the keyboard and do tasks faster.



Let's look at some of the most basic keyboard shortcuts for opening, shutting, and navigating through various slides.

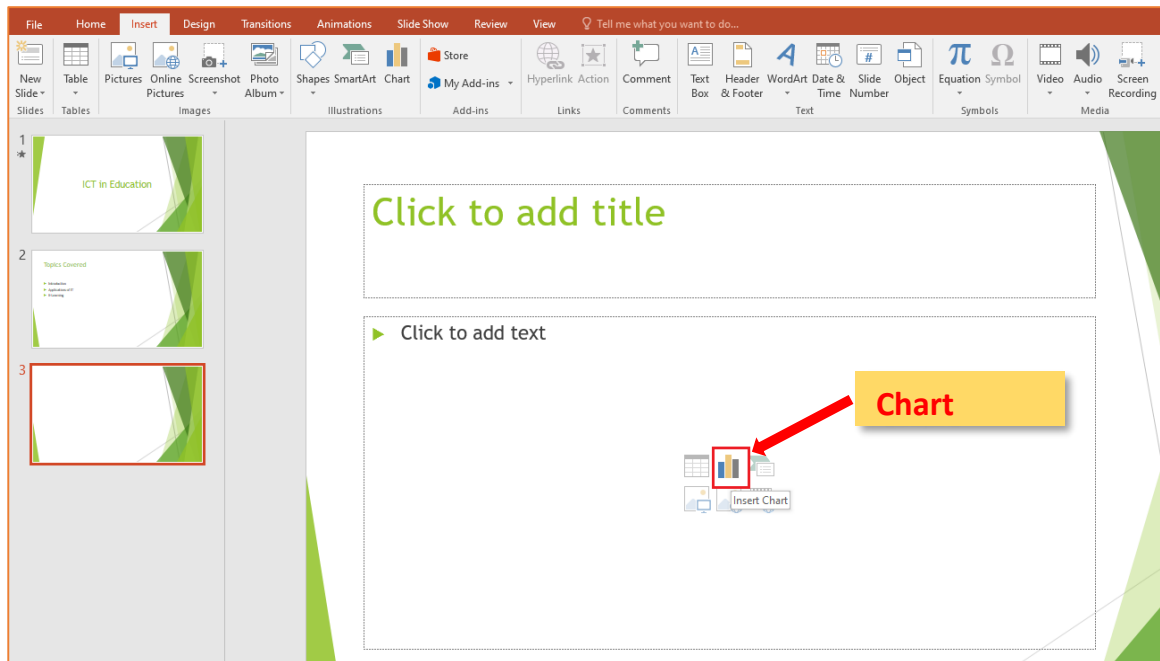
Shortcut Keys	Function
F5	Run a Slide Show
Esc	End a Slide Show
Ctrl + A	Select All
Ctrl + B	Bold
Ctrl + C	Copy
Ctrl + D	Duplicate Slide
Ctrl + E	Align Center
Ctrl + I	Italic
Ctrl + J	Justify
Ctrl + L	Align Left
Ctrl + M	Insert new slide
Ctrl + N	Create a New Presentation File
Ctrl + O	Open a previously saved presentation
Ctrl + P	Print a Presentation
Ctrl + Q	Save and close a presentation
Ctrl + R	Align Right
Ctrl + S	Save a presentation
Ctrl + U	Underline
Ctrl + V	Paste
Ctrl + Y	Redo
Ctrl + Z	Undo
Ctrl + Home	First Slide
Ctrl + End	Last Slide

INSERT CHART IN PRESENTATION

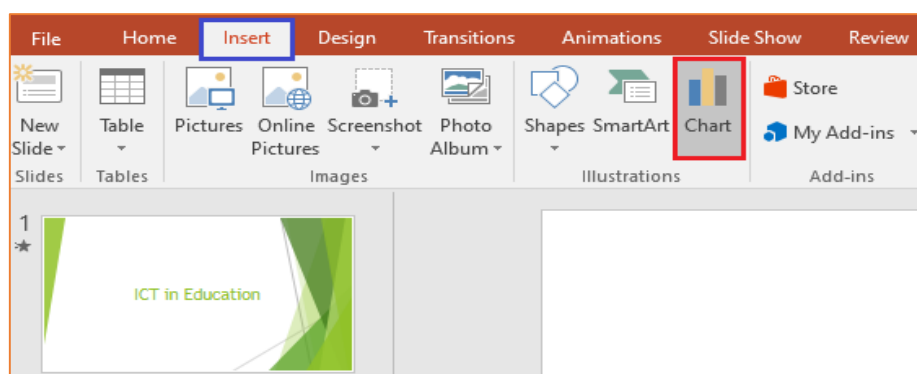
A chart is a graphical representation of data, in which the data is represented by symbols, such as bars in a bar chart or slices in a pie chart. Charts are used to display series of numeric data in a graphical format to make it easier to understand large quantities of data and the relationship between different series of data.

Inserting chart in a slide:

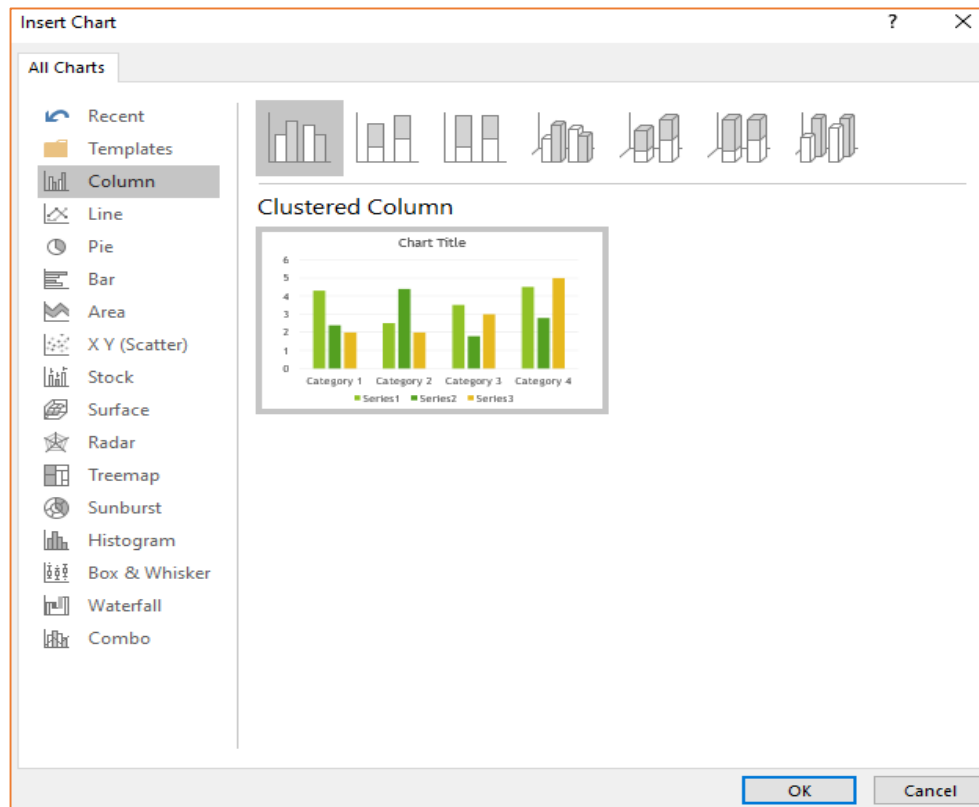
- Select the slide where you want to insert the chart.
- If your slide layout has a Content placeholder, click the **Chart** button.



- Alternatively, if you want to insert a chart in a slide that has no content placeholder, just select the **Insert** tab and click the **Chart** button.



- In the **Insert Chart** dialog box, select a chart based on your requirement and click the **OK** button.

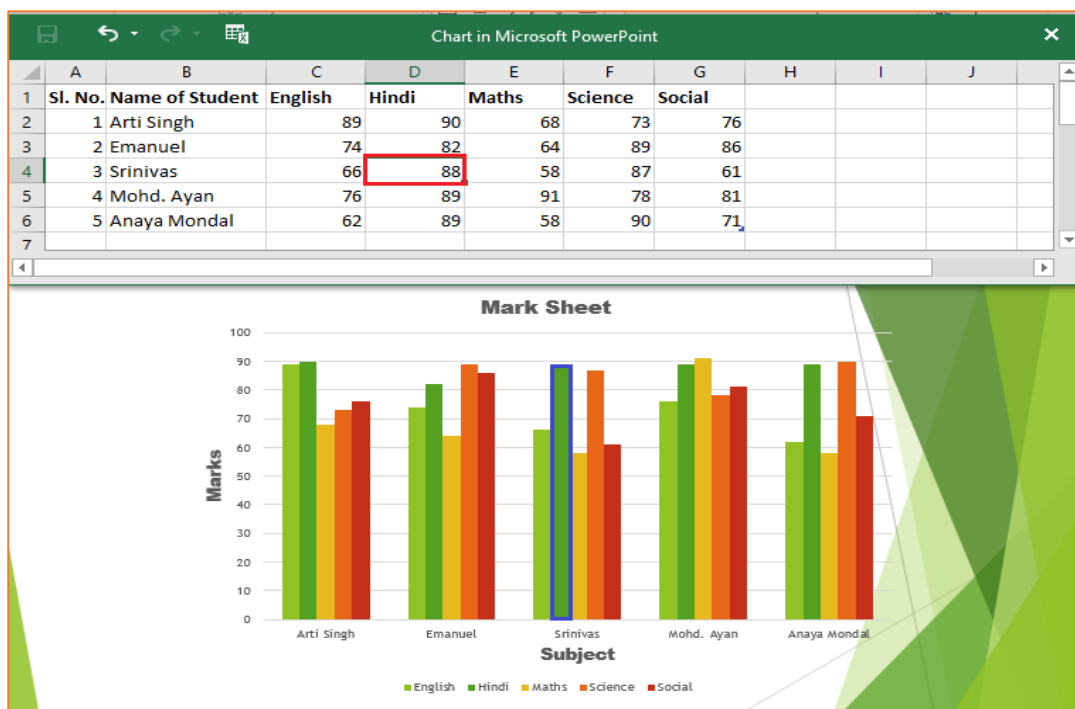


- A chart will be placed on the slide. An Excel sheet containing dummy data is also opened, if you want to see the larger Excel window, click the **Excel icon** within this sheet.

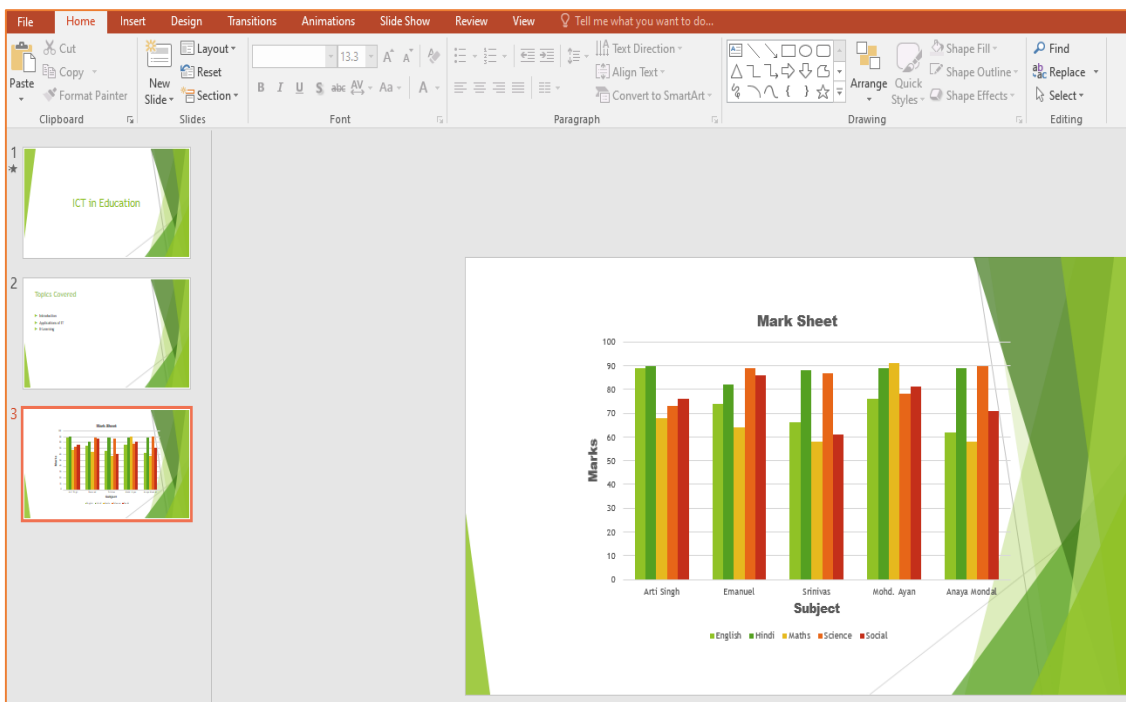


- Make changes as required. Any changes you make in the Excel table (highlighted in **red** in Figure) will reflect as changes on the chart within PowerPoint (highlighted in **blue** in Figure).

For example: To make chart for marks scored by students of Class 8 in all subjects.



➤ Close the Excel window. Now chart is placed on the slide.





1.key can be used to view Slide show.
2. In slides text is entered in.....
3. Word art and Picture options are available in Tab.
4. is the animation that occurs when you advance from slide to slide within the presentation.
5. To insert a new slide press

1.
2.
3.
4.
5.

1. What is the shortcut key to close PowerPoint?
a) Ctrl + Q b) F4 c) Ctrl + Delete
2. From which of these menus we can access a Text Box, Picture, Chart?
a) File b) Home c) Insert
3. Special effects used to animate objects of slides in slide show are called
a) Effects b) Custom animations c) Transitions



4. A chart can be put as a part of the presentation using
- a) Insert -> Chart b) Home -> Chart c) View -> Chart
5. Which of the following key will end the slide show?
- a) F5 b) Space bar c) Esc

Lab Activity

- I. Create a presentation on the topic “Types of Software” and do the following:
- a) Insert topic related images and word art
 - b) Apply transitions to the slides
 - c) Add custom animations



Learning Objectives

At the end of this chapter, the students will be able to -

- ✓ Define HTML
- ✓ List uses of HTML
- ✓ Use basic HTML tags

INTRODUCTION TO HTML

In today's world, the largest open network is the Internet, a worldwide computer network that provides access to different websites on World Wide Web (www).

You can download any information or documents containing text, sound, graphics or video from the Internet with the help of a web browser. In fact, **web browser** is software that enables a computer to display the documents located on computer's network. The documents on network are called **hypertext** (if the media is text only) and **hypermedia** (if it includes text as well as graphics).

Have you ever thought how these documents are displayed on the Internet? If your answer is no, then let us learn the procedure to view the information on computer screen.

The information appears page by page on the screen and these pages are referred as **web pages**. These contain text as well as graphics. These web documents are written in a special language called **HTML (Hyper Text Markup Language)**.

Hyper Text Markup Language (HTML) is a language to design and display web pages on the Internet with the help of web browsers. It includes texts as well as graphics.



- *HyperText* is the method by which you move around on the web by clicking on special text called hyperlinks which bring you to the next page.
- *Markup* is what HTML tags do to the text inside them. They mark it as a certain type of text for example italicized text.

BASIC STRUCTURE OF AN HTML DOCUMENT

An HTML basically comprises of text. The contents of the document are written using tags which are enclosed in angle brackets. These tags are understood by the web browser to display the contents of the document on the web page. However, tags don't appear on the web page.

The general format of an HTML document is given below:

```
<html>

  <head>
    <title> Title here </title>
  </head>

  <body>
    Web page content goes here.
  </body>

</html>
```

<html> </html>

This basically defines the document as web page. It also identifies the beginning and end of the HTML document. All other tags must fall between the html tags.

<head> </head>

The header contains information about the document that will not appear on the actual page, such as the title of the document, the author etc.

<title> </title>

The title tag defines the title that will appear in the title bar of your web browser. The title must appear between the head tags.



<body> </body>

The body tags contain all the information and other visible content on the page. All your images, links and plain text must go between the <body> and </body> tags.

EMPTY TAG AND CONTAINER TAG

HTML tags basically are of two types:

- Container tags
- Empty tags

Container tags:

These tags in an HTML document are in pairs (i.e., Start and end tags) along with angle brackets (<>). They are also known as ON tag and OFF tag respectively.

ON tag uses angle brackets whereas OFF tag is used with forward slash (/) after the opening of angle bracket (<).

<HTML>

.....
.....
.....
.....
.....
.....
.....
.....
.....

</HTML>

Empty tags:

Empty tags do not require a closing tag in an HTML document. It means, an empty tag has ON tag only in the document which is understood by the web browser.

Eg.,
 - to add a line break

<hr> - to add a horizontal line

 - to display image

<HTML>

<body>

<h1>Hello everyone!!! </h1>

**<p>This is a
 paragraph with a line break.</p>**

</body>

</HTML>



CREATION OF A WEBPAGE

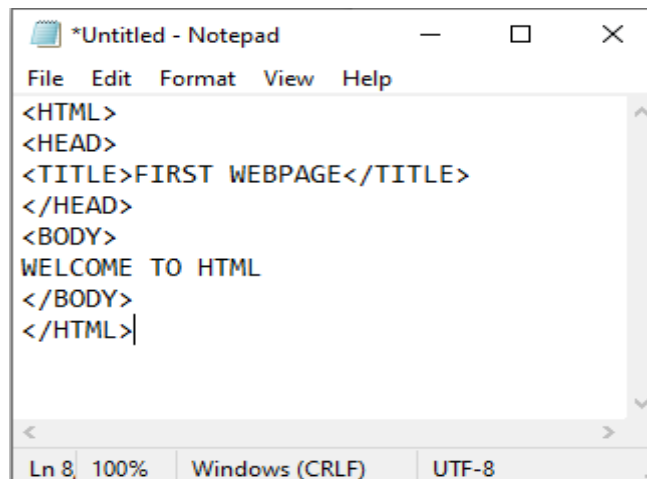
STEP 1: Open a text editor.

Start → All programs → Accessories → Notepad

OR

Type “Notepad” in the Search Box of Taskbar

STEP 2: In the appearing Notepad, type the HTML document.

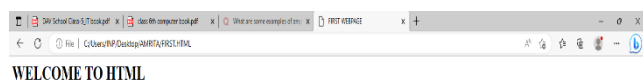


```
*Untitled - Notepad
File Edit Format View Help
<HTML>
<HEAD>
<TITLE>FIRST WEBPAGE</TITLE>
</HEAD>
<BODY>
WELCOME TO HTML
</BODY>
</HTML>
```

STEP 3: Save the file as .html or .htm

Click : File → Save (or) Press Ctrl + S → In the “File name” text box, type a file name “First.htm” with .htm or .html extension → Click the “Save” Button.

TO VIEW THE WEBPAGE IN THE BROWSER



STEP 1: Open a browser (Internet Explorer or Google Chrome or any browser)

STEP 2: File Open or press Ctrl+O

STEP 3: From the “Open” Dialog box, browse the folder in which the HTML document is saved.

Choose the file name and click “Open” button.

Now, your webpage is displayed in the browser as given in the figure:



Test your knowledge

I) FILL IN THE BLANKS:

1. HTML means
2. The text documents on internet are called
3. Every HTML document begins with and ends with tags.
4. The tags which comprise start and end tags, are called
5. Extension of a HTML file is.....
6. The tag must appear between the head tags.
7. Thetags contain all the information and other visible content on the page.
8. A.....tag doesn't have a closing end.

II) LAB ASSIGNMENT / PROJECT

Create different webpages to perform the following:

1. To display your name, class, father's name, mother's name etc.
2. To print the number from 1 to 15 on the screen.
3. To set the title of the page to "MY FIRST WEBPAGE" and display the body message as "WELCOME TO HTML WEB DESIGNING".

III) MULTIPLE CHOICE QUESTIONS:

1. HTML stands for
 - (a) Hyper Text Makeup Language
 - (b) Hyper Text Markup Language
 - (c) Hyper Text Machine Language
 - (d) Heavy Text Making Language
2. HTML documents include
 - (a) Text
 - (b) Graphics
 - (c) Sound
 - (d) all of the given
3. HTML file can be edited with
 - (a) Windows notepad
 - (b) MS-Word
 - (c) Word Perfect
 - (d) all of the given



UNIT 6

NETWORK, INTERNET & ITS APPLICATION

Learning Objectives

At the end of this chapter, the students will be able to -

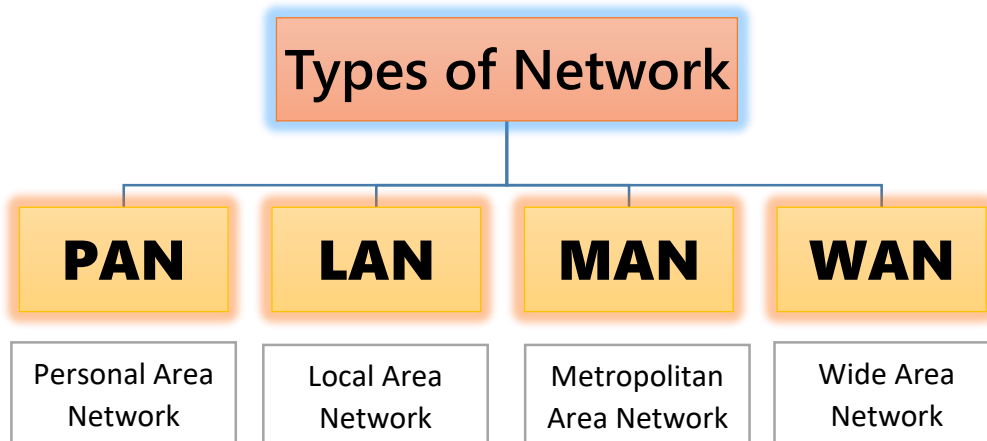
- ✓ Define what is Network & its types
- ✓ Understand Network Topology
- ✓ List the Advantages of Network
- ✓ Know about various Internet Services
- ✓ How to Access the Internet

INTRODUCTION TO COMPUTER NETWORK

A computer network is a cluster of computers over a shared communication path that works for the purpose of sharing resources from one computer to another, provided by or located on the network nodes.

Some of the uses of computer networks are the following:

- Communicating using email, video, instant messaging, etc.
- Sharing devices such as printers, scanners, etc.
- Sharing files
- Sharing software and operating programs on remote systems
- Allowing network users to easily access and maintain information



Personal Area Network (PAN)

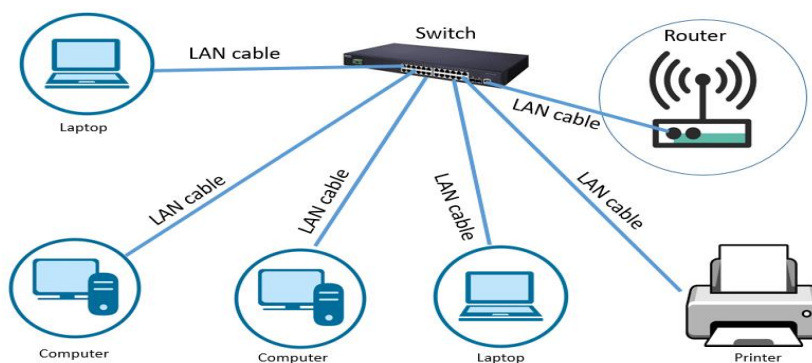
PAN is the most basic type of computer network. This network is restrained to a single person, that is, communication between the computer devices is centered only to an individual's workspace. Its transmission speed is very high with very



easy maintenance and very low cost. Examples of PAN are USB, computer, phone, tablet, printer, PDA, etc.

Local Area Network (LAN)

LAN is the most frequently used network. A LAN is a computer network that connects computers together through a common communication path, contained

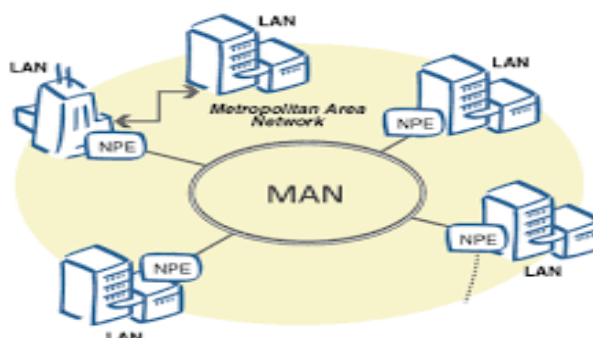


Local Area Network

within a limited area, that is, locally. It ranges up to 2km & transmission speed is very high with easy maintenance and low cost. Examples of LAN are networking in a home, school, library, laboratory, college, office, etc.

Metropolitan Area Network (MAN)

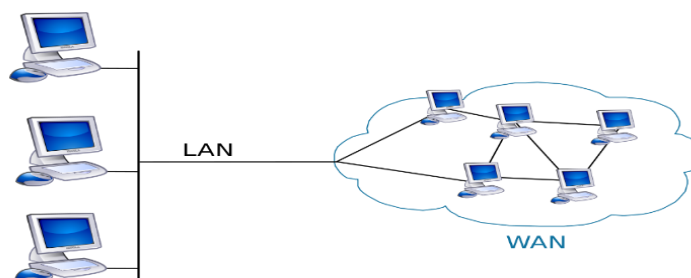
A MAN is larger than a LAN but smaller than a WAN. This is the type of computer network that connects computers over a geographical distance through a shared communication path



over a city, town, or metropolitan area. It is difficult to maintain and it comes with high cost. Examples of MAN are networking in towns, cities, a single large city, a large area within multiple buildings, etc.

Wide Area Network (WAN)

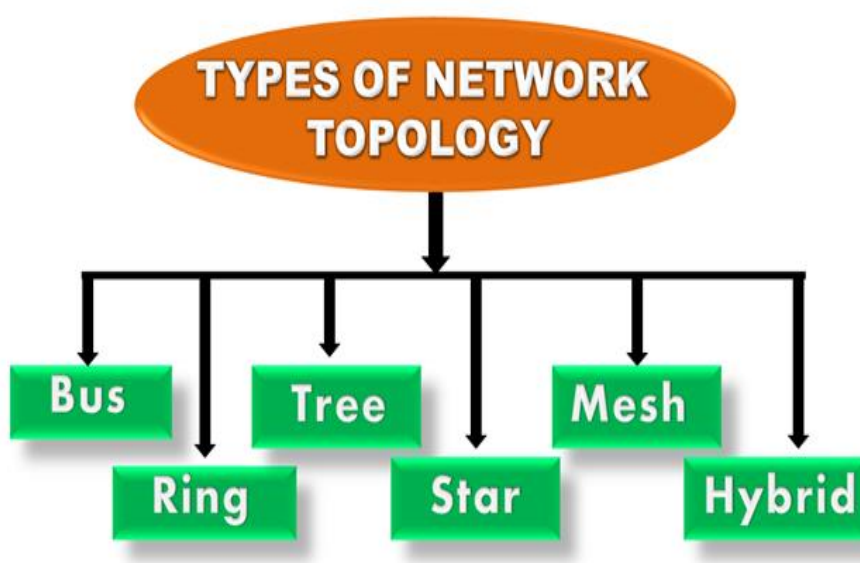
WAN is a type of computer network that connects computers over a large geographical distance through a shared communication path. It is not restrained to a single



location but extends over many locations. WAN can also be defined as a group of local area networks that communicate with each other with a range above 50km. The most common example of WAN is the Internet.

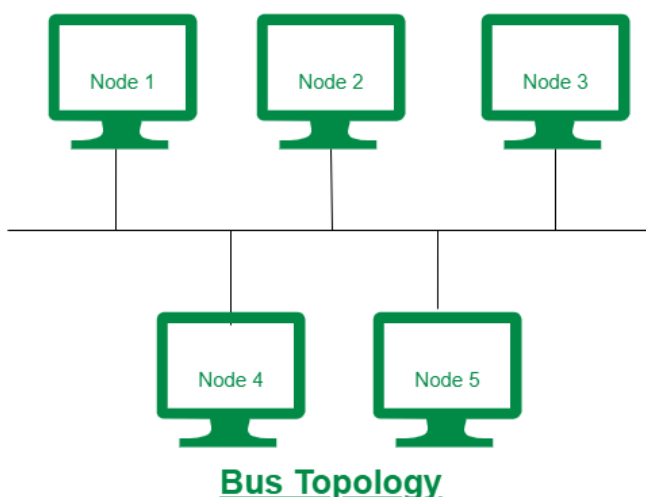
NETWORK TOPOLOGY

A network topology is the physical and logical arrangement of nodes and connections in a network. Nodes usually include devices such as switches, routers and software with switch and router features. Network topologies are often represented as a graph. Each type consists of different node and link configurations and has its own advantages and disadvantages.



Bus Topology

Bus topology is a network type in which every computer and network device is connected to a single cable. It is bi-directional. It is a multi-point connection and a non-robust topology because if the backbone fails the topology crashes. Alternatively called **line topology**.



BUS TOPOLOGY	
Advantages	Disadvantages
Cheap and easy to implement	Network disruption when computers are added or removed
Require less cable	A break in the cable will prevent all systems from accessing the network
Does not use any specialized network equipment	Difficult to troubleshoot

Ring Topology

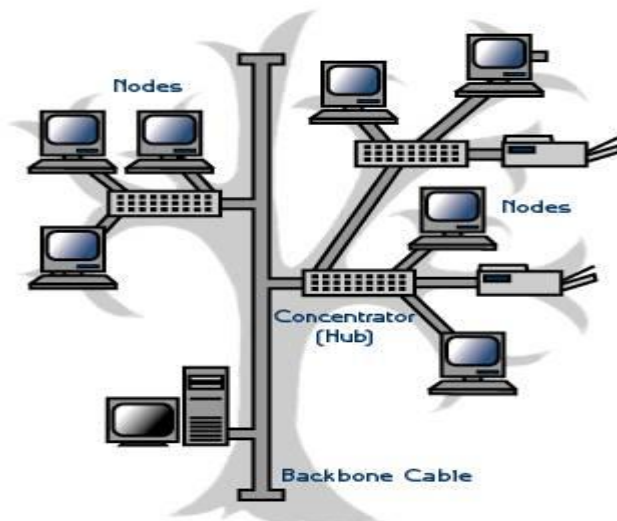
A ring topology is a network architecture in which devices are connected in a ring structure and send information to each other based on their ring node's neighbouring node. As

compared to the bus topology, a ring topology is highly efficient and can handle heavier loads. This forms a circular ring of connected devices which gives it its name. Data is commonly transferred in one direction along the ring, known as a unidirectional ring.



Tree Topology

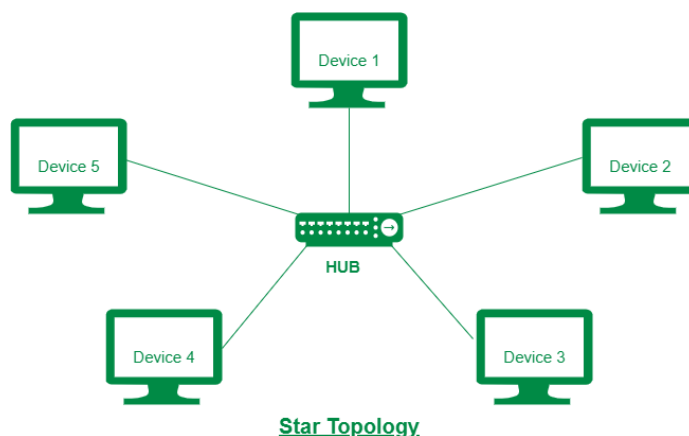
Tree topology is a sort of structure in which each node is related to the others in a hierarchy. In a topological hierarchy, there are at least three distinct levels. Sometimes it is also called hierarchical topology as in this topology, all elements are arranged like the branches of a tree. It is a lot like the star and bus topologies.



Tree topologies are commonly used to arrange data in databases and workstations in corporate networks. In a tree topology, any two linked nodes can only have one mutual connection, hence there can only be one link between them.

Star Topology

Star topology is a type of network topology in which every device in the network is individually connected to a central node, known as the switch or hub.



It is one of the most prevalent computer network configurations, and it's by far the most popular Network Topology. In this network arrangement, all devices linked to a central network device are displayed as a star. Here are a lot of examples of star topology in real life also, and you can see Star Network Topology Examples in Airports, Hospitals, Banks, and Educational Institutes. Data on a star network passes through the hub before continuing to its destination. The hub manages and controls all functions of the network. It also acts as a repeater for the data flow.

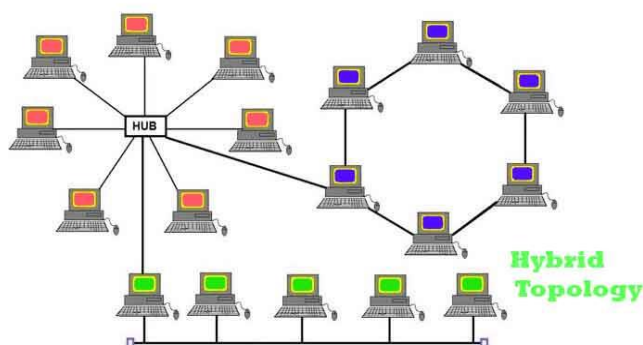
Mesh Topology

Mesh topology is a type of networking in which all the computers are inter-connected to each other. In Mesh Topology, the connections between devices take place randomly. The connected nodes can be computers, switches, hubs, or any other devices. In this topology setup, even if one of the connections goes down, it allows other nodes to be distributed.



Hybrid Topology

Hybrid topology is a type of network topology that uses two or more differing network topologies. These topologies can include a mix of bus topology, mesh topology, ring topology, star topology, and tree topology. The number of computers, their location, and desired network performance are all factors in the decision.



ADVANTAGES OF NETWORKS

1. Enhancement of Communication and Information Availability

Access to the web has made networking faster and more advanced in many aspects. The new technologies of messaging and sharing images, voices, and anything through textual messages have connected many users.

2. Convenient Sharing of Resources

In huge enterprises, there is a lot of daily functioning, resulting in data collection and generation, which can be a valuable resource for present or future endeavors. A computer network facilitates such organizations to share resources easily and at a faster rate to avoid any delay or drawback in



organizational work. It is one of the most important features of the computer network

3. Easy File/Data Sharing

The sharing of files and data sets is the most vital feature of a computer network. Many industries and organizations are using computer networks for their operations, and computer networks have made information sharing easier because such organizations run on their data. Computer networks also save a lot of time, and it has been proven to be monetarily beneficial in the long run.

4. Highly Flexible

The flexible nature of computer networks allows anyone worldwide to share information or data. This feature of computer networks has revolutionized the world and has made everything easier and more convenient.

5. Affordable

The software and networking devices are available at very affordable rates and provide easy ways to transfer information across the network.

6. Increases Cost Efficiency

Computer developers and engineers are constantly brainstorming and developing new kinds of software to enhance networking. There are a lot of websites on the Internet that even provide software that are free of cost and can also be used for multiple uses. The Internet, the primary source of networking in the world, is also home to many applications and software that are easily accessible, provide benefits for a long time, and can help with multiple things. Examples of such software that provide cost efficiency are VLC Media Player, Discord, WinRAR, Torrent, etc.

HISTORY OF INTERNET

We can track the origin of the internet back to the 1960s. In the late 1960s, the first-ever internet called the ARPANET (Advanced Research Projects Agency Network) was created. This network adopted the method of packet switching to allow multiple computers to communicate across a single network. This project was led by Robert Taylor and was proposed by Davies and Barn.



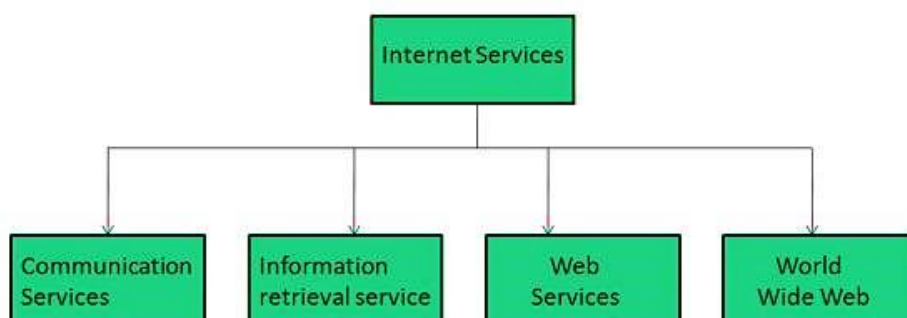
Advantages of Internet

The internet has become a household name since its easy installation and setup have been introduced. It was first invented for only government and military uses only. But now, it is installed in every house across the world.

1. It has increased the connectivity and the medium of sharing. Earlier it used to take days to send a single letter, but now our loved ones are just a few seconds away.
2. Banking has now become easier. Since the introduction of e-banking platforms, long tiresome waiting lines are a past. One can now communicate with its bank, transfer money, receive funds, etc., online.
3. E-commerce websites are another great addition to the advantages. One can buy clothes, groceries, home items, and much more online.
4. Internet is also a great source of entertainment. Platforms like Torrent, YouTube, Cloud Vibe, Stream, etc. are a great source of movies, videos, music, and games.
5. Social media platforms like Facebook, Instagram, Twitter have brought the world closer. We can stay connected to our friends and family 24×7 through them.
6. The education system has also revolutionized. Any student across the world can attend online classes. Physical presence in the class is no longer a necessity. This has boosted the literacy rate in third-world countries.

INTERNET SERVICES

Internet Services allows us to access huge amount of information such as text, graphics, sound and software over the internet. Following diagram shows the four different categories of Internet Services.





Communication Services

There are various Communication Services available that offer exchange of information with individuals or groups. The following table gives a brief introduction to these services:

S.No.	Service Description
1	Electronic Mail Used to send electronic message over the internet.
2	Telnet Used to log on to a remote computer that is attached to internet.
3	Newsgroup Offers a forum for people to discuss topics of common interests.
4	Internet Relay Chat (IRC) Allows the people from all over the world to communicate in real time.
5	Mailing Lists Used to organize group of internet users to share common information through e-mail.
6	Internet Telephony (VoIP) Allows the internet users to talk across internet to any PC equipped to receive the call.
7	Instant Messaging Offers real time chat between individuals and group of people. Eg. Yahoo messenger, MSN messenger.

Information Retrieval Services

There exist several Information retrieval services offering easy access to information present on the internet. The following table gives a brief introduction to these services:

S.N.	Service Description
1	File Transfer Protocol (FTP) Enable the users to transfer files.
2	Archie It's updated database of public FTP sites and their content. It helps to search a file by its name.
3	Gopher Used to search, retrieve, and display documents on remote sites.
4	Very Easy Rodent Oriented Net wide Index to Computer Achieved (VERONICA) VERONICA is gopher based resource. It allows access to the information resource stored on gopher's servers.

Web Services

Web services allow exchange of information between applications on the web. Using web services, applications can easily interact with each other. The web services are offered using concept of **Utility Computing**.

World Wide Web (WWW)

WWW is also known as W3. It offers a way to access documents spread over the several servers over the internet. These documents may contain texts, graphics, audio, video, hyperlinks. The hyperlinks allow the users to navigate between the documents.

Video Conferencing

Video conferencing or Video teleconferencing is a method of communicating by two-way video and audio transmission with help of telecommunication technologies.

Modes of Video Conferencing

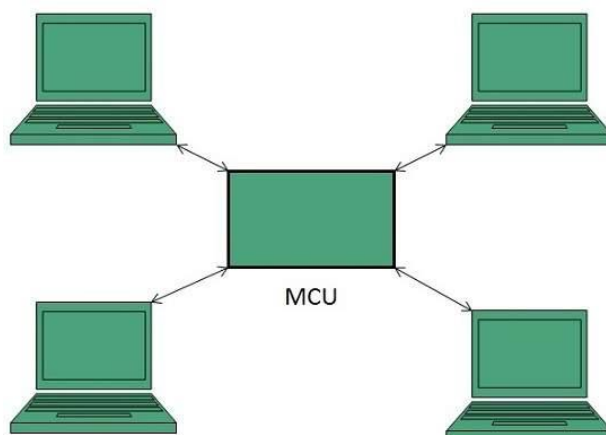
Point-to-Point

This mode of conferencing connects two locations only.



Multi-point

This mode of conferencing connects more than two locations through **Multi-Point Control Unit (MCU)**.





ACCESSING THE INTERNET

There are three elements needed to access the Internet from a laptop or desktop computer:

- (1) an ISP (2) a modem and (3) a Web browser.

The Internet Service Provider (ISP)

Access to the Internet is through an Internet service provider (ISP), which can be a large company such as Comcast or AT&T, or any of hundreds of smaller ISPs throughout the country.

The Modem

Depending on the kind of service you have, you will need a unit of hardware called a "modem" for connection. Today, cable and telephone companies provide service to most people and the modem converts their signals to the Internet packets your computer requires.

Rural areas may have only satellite service or dial-up telephone access, the latter as much as 100 times slower than cable. Older computers had built-in dial-up telephone ports.

Browsing the Web

A Windows PC comes with the Edge Web browser, while the Mac comes with Safari. Windows and Mac users quite often choose a different browser such as Firefox (www.mozilla.org) or Chrome (www.google.com/chrome), which offer additional features and are very popular.

Video conferencing tools

Video conferencing software is a software that allows two or more people to emulate a person-to-person meeting over the internet using real-time, multidirectional video and audio streaming.

Zoom.Google | Hangouts | Dialpad Meetings | TrueConf Online | Skype.
Free Conference | Lifesize Go | Slack Video Calls.

MOBILE APPS FOR STUDENTS

e-Pathshala

e-Pathshala, a joint initiative of Ministry of Education(MoE), Govt. of India and National Council of Educational Research and Training (NCERT) has been developed for showcasing and disseminating all educational e-resources including textbooks, audio, video, periodicals, and a variety of other digital resources.



e-Basta

e-Basta is a framework to make school books accessible in digital form as e-books to be read and used on tablets and laptops. The main idea is to bring various publishers (free as well as commercial) and schools together on one platform.

Digi Locker

DigiLocker is a flagship initiative of Ministry of Electronics & IT (MeitY) under Digital India programme. DigiLocker aims at 'Digital Empowerment' of citizen by providing access to authentic digital documents to citizen's digital document wallet. DigiLocker is a secure cloud based platform for storage, sharing and verification of documents & certificates.



DIKSHA (Digital Infrastructure for Knowledge Sharing): e-Learning



1. As part of PMeVidya announced under the Atma Nirbhar Bharat programme, DIKSHA is the 'one nation; one digital platform' for school education in India.
2. It was launched in 2017
3. It is a national platform available for schools in all states



4. DIKSHA is available for grades from 1 to 12.
5. DIKSHA can be accessed through mobile application and web portal.

(Digital Infrastructure for Knowledge Sharing) – What Does it Contain?

1. Courses for Teachers, quizzes and others
2. Lots of e-content will be provided which are aligned with the curriculum. Large number of case studies and solutions will be provided with the help of Energized Textbooks (ETB's) which are QR coded.
3. VidyaDaan was launched in April, 2020.
4. It is a content contribution program at national level, that makes use of the DIKSHA platform and tools, it allows donation or contribution of e-learning resources for school education by experts, private bodies, and educational bodies.

SELF ASSESSMENT

Multiple Choice Questions

1. Bluetooth is an example of
 - a) Wide area network
 - b) Virtual private network
 - c) Local area network
 - d) Personal area network
2. Which network topology requires a central controller or hub?
 - a) Star b) Mesh
 - c) Ring d) Bus
3. Data communication system within a building or campus is
 - a) LAN b) WAN
 - c) MAN d) PAN
4. WAN stands for
 - a) World area network c) Web area network
 - b) Wide area network d) Web access network



UNIT 7

CYBER SAFETY & SECURITY - II



Learning Objectives

In this chapter, you will be able to understand:

- ✓ What is Cyber Safety and digital footprints
- ✓ Cyber Etiquettes- Do's & Don'ts while using social media and other internet services
- ✓ Basic cyber safety guidelines
- ✓ Cyber Law- Indian IT Act

Cyber Safety

Cyber Safety is a measure of promoting the safe usage of the internet, especially in order to protect against viruses, fraud, or other forms of attack.

Advantages of Cyber Safety

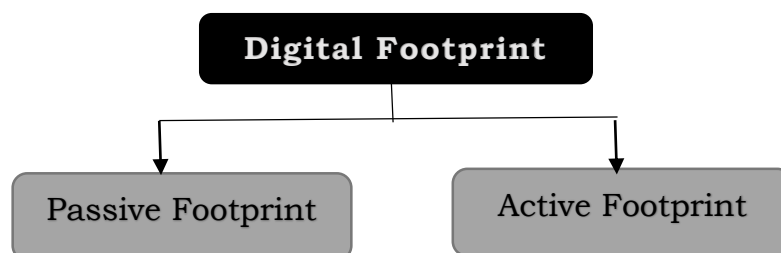
- ✓ Protects sensitive personal information
- ✓ Protect private and Business data
- ✓ Improve the overall security of an organisation
- ✓ Improves productivity

Digital footprints

A digital footprint is data that is left behind when users have been online.



There are two types of digital footprints which are **passive** and **active**.



A **Passive Footprint** is made when information is collected from the user without the person knowing this is happening.

An **Active Footprint** is where the user has deliberately shared information about themselves either by using social media sites or by using websites.

Actions which increase the size of Digital Footprint

- Making purchases online
- Opening a credit card account
- Using a mobile banking app
- Buying or selling stocks
- Registering your email address with a gym
- Subscribing to a health and fitness blog
- Using apps to track your activities and workouts
- Subscribing to an online news source
- Viewing articles on a news app
- Using social media on your computer or devices
- Logging into sites with your social media credentials



Sharing information, data, and photos with your connections

Cyber Etiquettes-Do's & Don'ts while using social media and other internet services



[Cyber Etiquettes](#)

<https://www.youtube.com/watch?v=XfMZDBiBDOI&list=PLZ1IjQtA2fL11yY06-HnpsrJifRsYoeFq&index=5>





<u>Do's</u>	<u>Don'ts</u>
Create strong passwords.	Never write down your password.
Change your password every ninety days.	Never give out your password to anyone.
Perform regular backups of important data.	Never select the "Remember My Password" option.
Create a password for your files in order to protect file sharing activities.	Never purchase anything promoted in a spam message.
Delete any message that refers to groups or organizations that you are not a part of.	Please refrain from opening an e-mail attachment, even from someone you know well, unless you were expecting it.
Download and install software only from online sources you trust.	Avoid creating common passwords such as your name, etc.
Never click on a link from an untrusted source.	Never reply to e-mail(s) requesting financial or personal information.
Close windows containing pop-up ads or unexpected warnings by clicking on the "X" button in the upper most right hand corner of that window, not by clicking within the window.	Avoid opening e-mail(s) or e-mail attachments from an unknown sender.
Use antivirus software, and update it on a regular basis to recognize the latest threats.	Please refrain from clicking on the close button within pop-up ads.
Regularly update your operating system, Web browser, and other major software, using the manufacturers' update features, preferably using the auto update functionality.	Under no circumstances should you install or use pirated copies of software.



Netiquette for Social Media

[Netiquette for Social Media](https://www.youtube.com/watch?v=a38MJVEPWiw)

<https://www.youtube.com/watch?v=a38MJVEPWiw>

Basic Cyber Safety Guidelines

- Report phishing issues to the concerned authorities or at cybercrime.gov.in.
- In case of unwanted mails, mark them as spam mails.
- Be cautious while making friends on the social networking site.
- Do not send or accept friendship requests from any unknown user.
- Do not post any unpleasant content on social networking site, it is a cyber-crime.
- Do not post and spread any kind of a rumour, it is a cyber-crime.
- If someone is harassing or threatening you, take snapshot of it as a proof, and block the person. Also, report the incident to the site administrator.
- Use updated antivirus and firewall for secure browsing and password management techniques.
- Make sure that the website address is properly spelled. Because there may be two websites with almost the same name, one being a phishing website.
- Delete cookies periodically along with the temporary files stored on your system during web browsing. A cookie is a small piece of information about the client browsing a website. By obtaining access to these cookies, hackers may gain unauthorized access to these websites.



Cyber Law- Indian IT Act

The Information Technology Act, 2000 also known as an IT Act is an act proposed by the Indian Parliament reported on 17th October 2000. The main objective of this act is to carry lawful and





trustworthy electronic, digital and online transactions and alleviate or reduce cybercrimes. Major amendments were made in 2008 to address issues like cyber terrorism, data protection, child pornography, stalking etc.

Sections and Punishments under IT Act, 2000 are as follows:

SECTION	PUNISHMENT
Section 43	This section of IT Act, 2000 states that any act of destroying, altering or stealing computer system/network or deleting data with malicious intentions without authorization from owner of the computer is liable for the payment to be made to owner as compensation for damages.
Section 43A	This section of IT Act, 2000 states that any corporate body dealing with sensitive information that fails to implement reasonable security practices causing loss of other person will also liable as convict for compensation to the affected party.
Section 66	Hacking of a Computer System with malicious intentions like fraud will be punished with 3 years imprisonment or the fine of Rs. 5,00,000 or both.
Section 66 B, C, D	Fraud or dishonesty using or transmitting information or identity theft is punishable with 3 years imprisonment or Rs. 1,00,000 fine or both.
Section 66 E	This Section is for Violation of privacy by transmitting image or private area is punishable with 3 years imprisonment or 2,00,000 fine or both.
Section 66 F	This Section is on Cyber Terrorism affecting unity, integrity, security, sovereignty of India through digital medium is liable for life imprisonment.
Section 67	This section states publishing obscene information or pornography or transmission of obscene content in public is liable for imprisonment up to 5 years or fine of Rs. 10,00,000 or both.



Activities

SELF-ASSESSMENT

Take this short quiz to help evaluate your screen time on social media

1. How much time do you usually spend on social media daily?

- A) Less than an hour
- B) 3-4 Hours
- C) 1-2 Hours
- D) 5-6 Hours

2. You've taken a photo. How long does it take you to edit and upload it?

- A) About a minute
- B) 5-10 minutes
- C) A couple of minutes
- D) Half an hour

3. When are you most often distracted by social media?

- A) Very rarely
- B) During conversations
- C) When I'm eating
- D) When I'm supposed to be doing work
- E) All the time

4. Do you feel your social media use is negatively impacting your relationships?

- A) Absolutely not
- B) I've been told off a couple of times
- C) I don't think so
- D) Only with certain people
- E) Yes, quite a bit

5. Has anyone close to you ever complained about your time on social media?

- A) Never
- B) Maybe once or twice
- C) My parents
- D) My teachers
- E) Yes, every time I go on my phone

6. Do you scroll through social media near bedtime or when you are in bed?

- A) Never
- B) I only answer text messages
- C) I have a little scroll
- D) I try to limit it to around 30 minutes
- E) Definitely!



7. How often do you check social media notifications?

- A) Once a day
- B) A couple of times throughout the day
- C) Every couple of hours
- D) Probably once an hour
- E) Every 10-20 minutes

8. When you do not have access to social media, how do you feel?

- A) It doesn't bother me
- B) Relaxed
- C) Bored
- D) Like I'm missing out
- E) Anxious

9. How would you feel if you got no likes on a post?

- A) I do not care
- B) Meh. It probably wasn't a great post!
- C) Must. Try. Harder.
- D) I'd ask friends to like it
- E) Embarrassed, I'd delete it



PROJECT WORK

Topic: Problem Solving, Scratch Programming, Presentation Tool & HTML

SUGGESTED ACTIVITIES:

1. Ask the student to write 02 Algorithms on any given topic and tell them to prepare the flowchart for the same.
2. Create a Scratch Project by adding a Sprite and adding various Blocks to move the sprite in a given direction. Ask the student to add sound and also change the backdrop, costume of the sprite.
3. Make a presentation on MS-Power Point with atleast 07 slides on any suggested topic using the following features:
 - a. Adding Word Art.
 - b. Adding Audio and Video in the slides
 - c. Use of Transitions and Custom Animations
 - d. Use of chart to show a data graphically.
4. Write a basic HTML program in notepad using tags like <Head>, <Body>, <Title> and saving the file. Run the program and view the same as a webpage in a browser.



PARAM is a series of supercomputers designed and assembled by the Centre for Development of Advanced Computing (C-DAC) in Pune, India. PARAM means "supreme" in the Sanskrit language, whilst also creating an acronym for "PARAllel Machine".



STATE INSTITUTE OF EDUCATION

DEPARTMENT OF EDUCATION
ANDAMAN & NICOBAR ISLANDS